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Computer Science

B.Sc. Graduation Project



AI-Powered Sensory Substitution System

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December – 2025



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ACKNOWLEDGEMENT

We would like to express our deep gratitude to the Arab Academy for Science and Technology for their support and guidance throughout this project. Their invaluable contribution has been a key factor in the success of our work. We are also extremely grateful to our supervisor Prof. Fahima Maghraby, for her valuable insights and guidance. Her expertise and support have been invaluable to us, and we are deeply thankful for her ongoing support and encouragement. Finally, we would like to express our gratitude to our country and government for providing the resources and support necessary to complete this project. We are proud to be able to contribute to the development and advancement of our nation through our work.

ABSTRACT

Blind and Low Vision (BLV) individuals face persistent challenges in navigating environments and interacting with physical objects due to the absence of visual information. While traditional mobility aids and existing assistive technologies provide partial support, most solutions focus on isolated tasks such as obstacle detection or text recognition and lack integrated, real-time environmental understanding. This project presents Echora, an AI-powered sensory substitution system designed to enhance independence, safety, and situational awareness for BLV users through multimodal perception and feedback.

Echora combines wearable sensing hardware with smartphone-based artificial intelligence to translate spatial and semantic information into intuitive audio and haptic cues. The system integrates depth sensing, computer vision and inertial measurement to support both macro-navigation (environmental awareness and obstacle avoidance) and micro-interaction (precise hand guidance and object interaction). By employing real-time multimodal sensor fusion and adaptive mode switching, Echora delivers context-aware feedback while minimizing cognitive load on the user.

All high-level processing and AI inference are performed locally on the mobile device, ensuring low latency, privacy preservation, and independence from cloud connectivity. The wearable components are limited to sensing and feedback, enabling a lightweight, modular, and scalable system architecture. Key features include continuous spatial audio navigation, haptic collision alerts, object and text recognition, and hand-guided interaction.

The proposed system is designed to complement, rather than replace, existing mobility aids such as white canes or guide dogs. By unifying navigation, interaction, and perception within a single adaptive framework, Echora demonstrates the potential of AI-driven sensory substitution systems to significantly improve autonomy and quality of life for Blind and Low Vision users.

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Chapter 1

Introduction

1.1 Introduction

Vision plays a fundamental role in human perception, enabling individuals to navigate environments, recognize objects, and interact safely with the physical world. For Blind and Low Vision (BLV) individuals, the absence or limitation of visual input significantly impacts mobility, independence, and confidence in daily activities. While traditional assistive tools such as white canes and guide dogs provide essential support, they offer limited environmental information and often rely on direct physical contact.

Recent advances in artificial intelligence (AI), machine learning, and wearable sensing technologies have enabled new forms of assistive systems that compensate for visual impairment through alternative sensory channels. Sensory substitution approaches transform visual and spatial information into auditory feedback, allowing users to perceive aspects of their surroundings without relying on vision. These systems aim to improve environmental awareness, enhance safety, and support independent interaction with objects and spaces.

Project **Echora** is proposed as an intelligent sensory substitution system that integrates AI-driven perception with multimodal sensing and real-time feedback. The system is designed to support both large-scale environmental navigation and fine-grained object interaction by translating spatial and semantic information into intuitive audio and haptic cues. By operating as an adaptive and context-aware digital sense, Echora seeks to move beyond single-purpose assistive devices toward a unified perceptual solution.

This document presents the requirements and design considerations of the Echora system. It defines the project purpose, scope, objectives, and constraints, and serves as a foundation for the system architecture, functional requirements, and implementation details described in subsequent sections.

1.2 Purpose

The purpose of this document is to define the requirements, scope, and constraints of the Echora project, an AI-powered sensory substitution system for Blind and Low Vision users. This document serves as a reference for stakeholders, developers, and evaluators by clearly specifying the system's intended functionality, operating environment, and design considerations.

Additionally, this document provides a structured foundation for the design, development, and evaluation phases of the project. By outlining both functional and non-functional requirements, it ensures that the system is developed in alignment with user needs, technical feasibility, and project objectives.

1.3 Intended Use

The Echora system is intended to assist Blind and Low Vision (BLV) users in navigating environments and interacting with physical objects in a safe and independent manner. The system is designed for everyday use in both indoor and outdoor settings, such as homes, workplaces, public buildings, and urban spaces.

Echora is expected to be worn and used continuously while the user is moving or performing tasks that require spatial awareness or object interaction. It provides real-time audio and haptic feedback to convey information about surroundings, obstacles, and points of interest, enabling users to make informed movement and interaction decisions without relying on visual input.

The system is intended to operate autonomously with minimal user intervention, adapting its behavior based on user context and activity. Echora is not designed to replace traditional mobility aids such as white canes or guide dogs but rather to complement them by providing additional environmental and interaction-related information.

1.4 Project Overview

Project Echora is an AI-powered sensory substitution system designed to assist Blind and Low Vision (BLV) users by translating environmental and object-related information into intuitive audio and haptic feedback. The system combines wearable sensing hardware with smartphone-based artificial intelligence to provide real-time perception of the user's surroundings.

Echora operates through a multimodal approach that supports both large-scale environmental awareness and fine-grained object interaction. By integrating depth sensing, computer vision, active acoustic sensing, and haptic feedback, the system enables users to navigate spaces, locate objects, and interact with them more safely and confidently. The system is designed to adapt its feedback based on user context, allowing seamless transitions between navigation and interaction tasks.

The project emphasizes modularity and scalability, with computation performed on a mobile device to leverage modern AI capabilities while keeping wearable hardware lightweight. Echora is developed as an assistive system that complements existing mobility aids and aims to enhance user independence through a unified digital perception of the environment.

1.5 Problem Statement

Blind and Low Vision (BLV) individuals face significant challenges in navigating environments and interacting with physical objects due to the lack of visual information. While traditional mobility aids such as white canes and guide dogs provide essential support, they primarily rely on direct physical contact and offer limited awareness of the surrounding environment beyond immediate obstacles. As a result, users often lack advance information about spatial layout, object location, and object properties, which can lead to inefficiency, safety risks, and reduced independence.

Existing assistive technologies typically address isolated tasks, such as text recognition, basic obstacle detection, or audio-based scene description. These solutions often fail to provide continuous spatial awareness, precise guidance for object interaction, or an integrated understanding of the environment. Furthermore, most current systems do not support pre-contact perception of objects, forcing users to rely on trial-and-error touch when interacting with unfamiliar items.

Therefore, there is a need for an assistive system that can provide real-time, integrated spatial and semantic information to BLV users. Such a system should support both navigation and fine-grained object interaction while operating in everyday environments with minimal user

effort and without replacing existing mobility aids.

1.6 Motivation

The motivation for developing the Echora system is grounded in the growing global and local impact of vision impairment and the limitations of existing assistive technologies in addressing this challenge. Vision loss remains a major public health and socioeconomic issue, affecting hundreds of millions of individuals worldwide and significantly reducing independence, safety, and quality of life for Blind and Low Vision (BLV) users.

As illustrated in Figure 1.1, global estimates indicate that billions of people are affected by various eye conditions that lead to vision impairment, highlighting the scale of the problem on a worldwide level. This global burden is further reflected at the regional level, where Figure 1.2 demonstrates the substantial BLV population across both worldwide and Middle Eastern contexts. These figures emphasize that vision impairment is not a niche concern but a widespread issue requiring scalable and accessible technological solutions.

The situation is particularly critical at the national level. Figure 1.3 presents the distribution of the BLV population in Egypt, revealing a significant number of individuals affected by blindness and moderate to severe vision loss. Moreover, the trend shown in Figure 1.4 indicates a continuous increase in sight loss over time, suggesting that the challenge is expected to grow rather than diminish in the coming years. This rising trend places additional pressure on healthcare systems, families, and support infrastructures.

Beyond personal and social challenges, vision impairment also imposes a substantial economic burden. As shown in Figure 1.5, productivity loss due to poor vision in Egypt results in considerable annual economic costs. These losses stem from reduced workforce participation, decreased efficiency, and increased dependency, reinforcing the need for solutions that enhance independence and functional ability for BLV individuals.

Despite these challenges, many existing assistive technologies provide only partial support, often focusing on isolated tasks such as obstacle detection or text reading. Such solutions frequently lack integrated spatial awareness, real-time feedback, and support for fine-grained object interaction, limiting their effectiveness in complex real-world environments.

Echora is motivated by the need to address these gaps through an integrated, AI-powered sensory substitution system. By leveraging advances in artificial intelligence, wearable sensing, and mobile computing, Echora aims to translate spatial and semantic information into intuitive audio and haptic feedback. This approach seeks to reduce cognitive load, improve safety, and empower BLV users with a more comprehensive and proactive understanding of their surroundings, directly responding to the scale, trends, and impacts highlighted in Figures 1.1 through 1.5.

Figure 2.1 Global estimates of numbers of people affected by selected eye conditions that can cause vision impairment

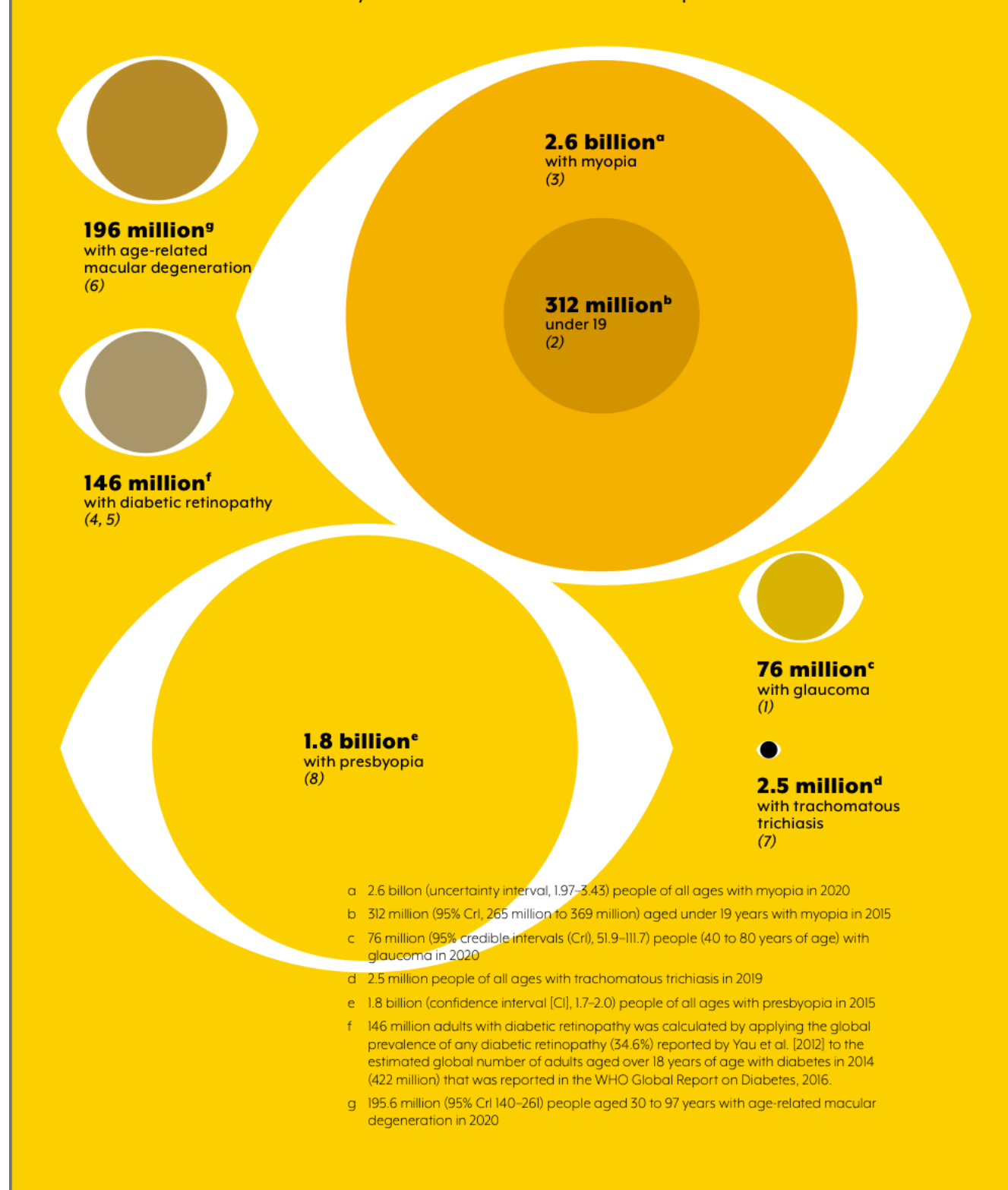
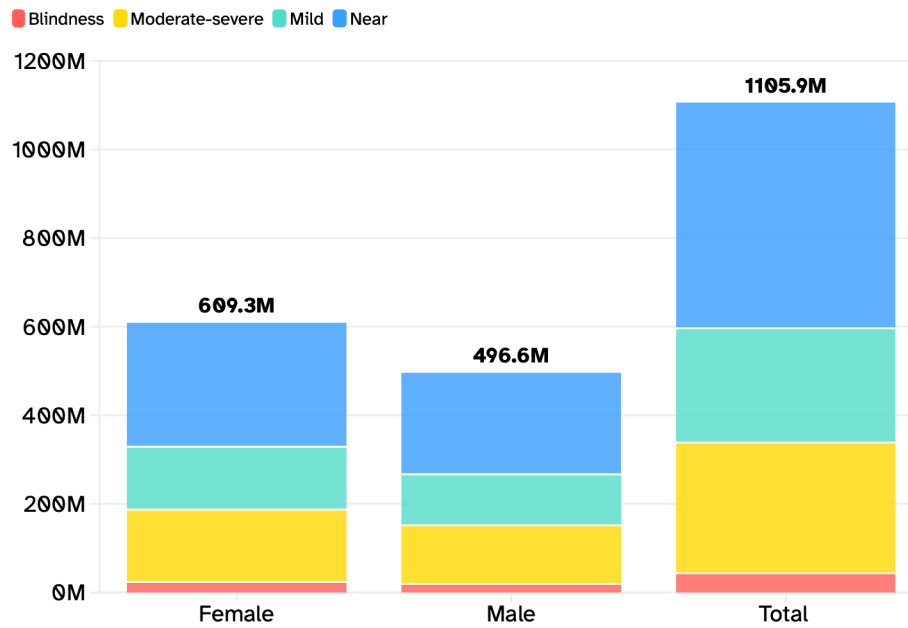
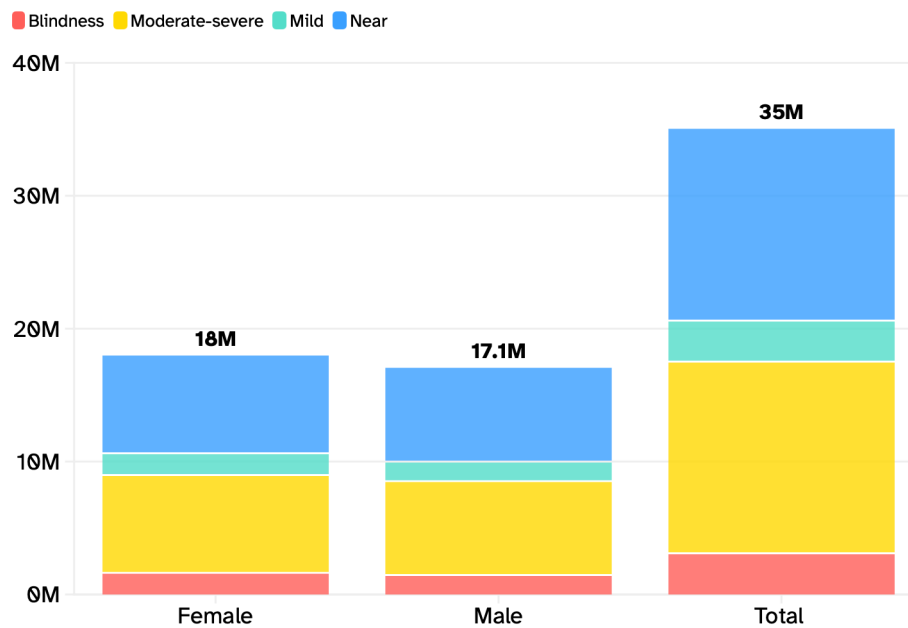


Figure 1.1: Global estimates of the number of people affected by major eye conditions causing vision impairment



How to cite: Bourne R, et al. Trends in prevalence of blindness and distance and near vision impairment over 30 years: an analysis for the Global Burden of Disease Study. Lancet Glob Health. 2020. Accessed via the IAPB Vision Atlas: visionatlas.iapb.org.



How to cite: Bourne R, et al. Trends in prevalence of blindness and distance and near vision impairment over 30 years: an analysis for the Global Burden of Disease Study. Lancet Glob Health. 2020. Accessed via the IAPB Vision Atlas: visionatlas.iapb.org.

Figure 1.2: Numbers of BLV population Worldwide and Middle East

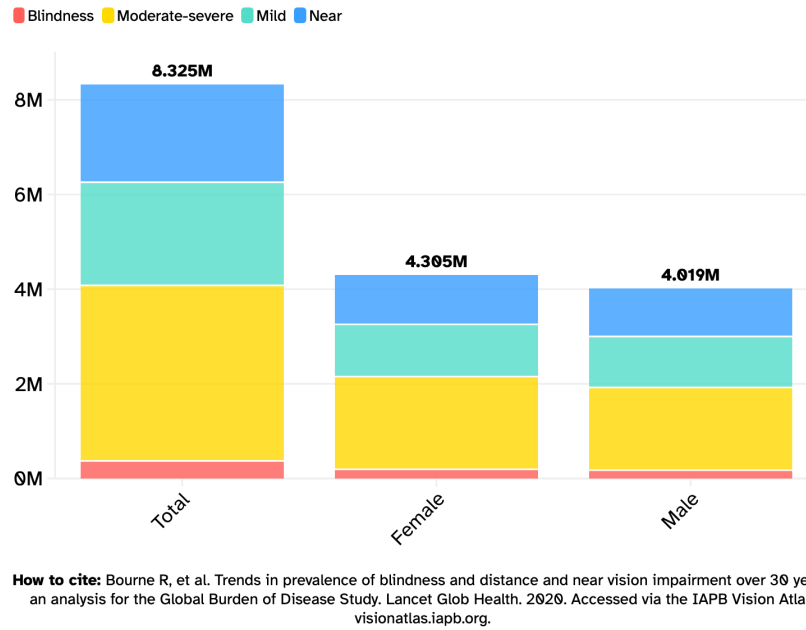


Figure 1.3: Numbers of BLV population in Egypt

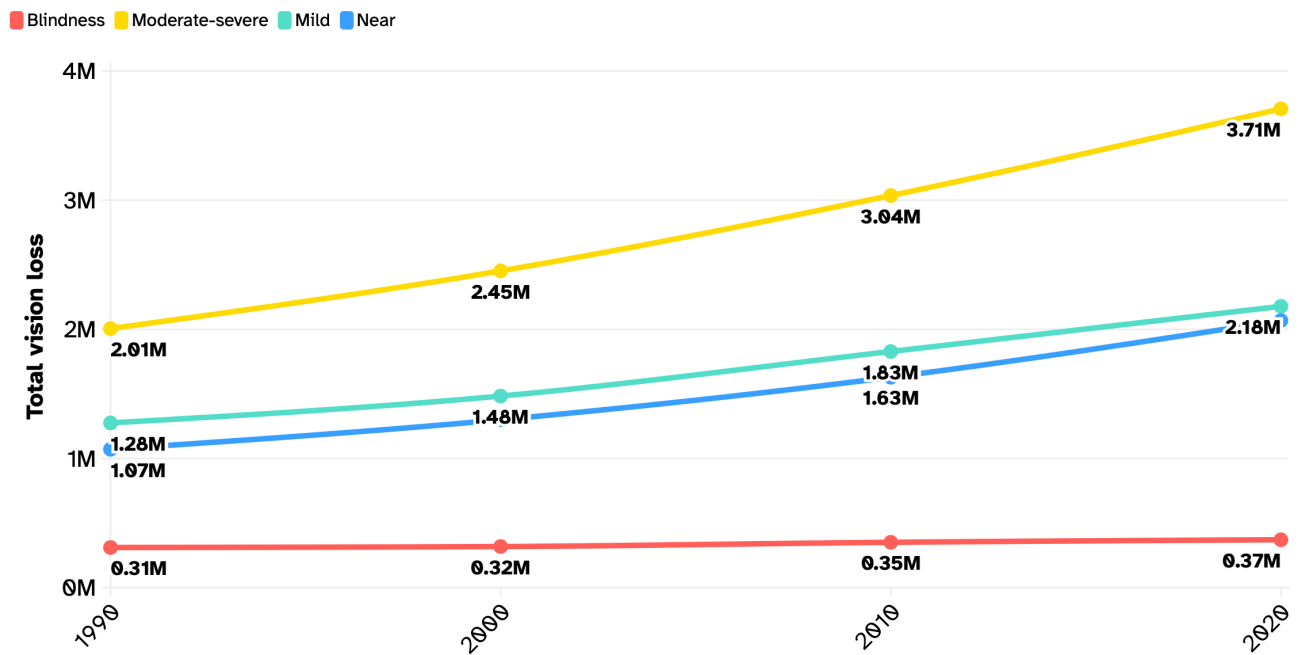
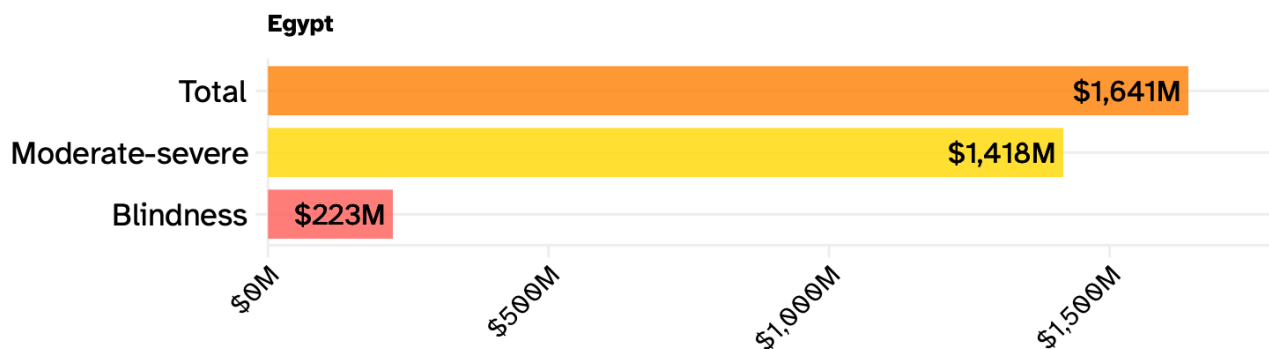


Figure 1.4: Sight loss over time in Egypt

Blindness Moderate-severe Total



How to cite: Eckert KA, et al. Update of a simple model to calculate the annual global productivity loss due to blindness and moderate and severe vision impairment. *Ophthalmic Epidemiol.* 2022;1-9. Accessed via the IAPB Vision Atlas: visionatlas.iapb.org.

Note: Median estimate (calculated), GNI Method.

Figure 1.5: Productivity loss due to poor vision per year in Egypt

1.7 Document Conventions

This document follows a set of conventions to ensure clarity, consistency, and ease of understanding throughout the specification. These conventions define how terminology, requirements, and formatting are presented to maintain a professional and structured technical document.

1.7.1 Typography Conventions

Bold text is used to highlight key terms, feature names, and section headings upon their first occurrence in the document. Italic text is applied for emphasis and for introducing important concepts that require special attention. Monospace text is reserved for technical commands, code references, system identifiers, and system-generated messages. Functional requirements are numbered using the format FRX.Y, where X represents the feature number and Y represents the specific requirement within that feature.

1.7.2 Requirement Priority Levels

Requirements in this document are categorized based on their importance to system functionality and user safety. Critical (Live) requirements represent essential features that must be implemented to ensure core system operation and protect the user. High-priority requirements significantly enhance system performance and user experience but are not strictly mandatory for basic operation. Medium-priority requirements provide extended or optional functionality that may be included in later development stages.

1.7.3 Stimulus-Response Format

Functional requirements are specified using a stimulus-response format to clearly define system behavior. Each requirement describes a stimulus, which represents an event, input, or condition that triggers system behavior, followed by a response, which defines the expected system output or action resulting from that stimulus. This format supports clear validation and traceability of system behavior.

1.8 Intended Audience

This document is intended for multiple stakeholder groups involved in the development, evaluation, and understanding of the Echora system.

1.8.1 Primary Audience

The primary audience includes project developers and system designers responsible for implementing the Echora system, as well as testers and evaluators tasked with validating functional and non-functional requirements. It also serves as a reference for team members throughout the development lifecycle to ensure alignment with project objectives and specifications.

1.8.2 Secondary Audience

The secondary audience consists of course instructors and academic supervisors overseeing the project, in addition to reviewers and evaluators assessing the project's feasibility, technical correctness, and scope.

1.8.3 Tertiary Audience

The tertiary audience includes researchers and students interested in assistive technologies and sensory substitution systems, accessibility advocates, potential collaborators seeking an overview of the project, and readers with a general interest in AI-powered wearable assistive solutions.

1.9 Project Scope

This section defines the scope of the Echora project by outlining system boundaries, architectural assumptions, included features, excluded items, and supported Sustainable Development Goals.

1.9.1 System Architecture

The Echora system follows a distributed architecture that separates sensing and actuation from computation and intelligence. The architecture is composed of two main components: wearable hardware and smartphone-based software.

The wearable system consists of smart glasses equipped with a depth camera for environmental sensing, an inertial measurement unit for motion and orientation tracking, haptic actuators positioned on the wrist to deliver tactile feedback, a low-power microcontroller responsible for sensor data acquisition and communication, and a rechargeable battery that powers the wearable components. These hardware elements are designed exclusively for sensing and feedback purposes, with no artificial intelligence or high-level decision-making performed on the wearable device itself.

The smartphone-based software serves as the central processing unit of the system. It performs real-time processing of depth and visual data, executes artificial intelligence models for spatial perception, object detection, hand tracking, and semantic understanding, manages navigation logic and automatic mode switching between navigation and interaction, generates spatial audio feedback, and issues haptic control commands to the wearable device. This separation allows the wearable hardware to remain lightweight while enabling scalable software upgrades without requiring hardware modifications.

1.9.2 Communication Protocol

The Echora system employs wireless communication to support real-time data exchange between the wearable hardware and the smartphone software. The communication design prioritizes low latency, reliability, and power efficiency to support continuous operation.

A combination of Wi-Fi and Bluetooth Low Energy (BLE) is used to achieve this balance. Wi-Fi communication is utilized for transmitting high-bandwidth data such as depth frames and visual information, while BLE is used for low-bandwidth, low-power data exchange including sensor readings, control signals, and haptic commands. The wearable device transmits raw sensor data, including depth measurements and inertial data, to the smartphone for processing. In return, the smartphone sends control instructions to the wearable device, such as haptic feedback commands, mode changes, and warning signals. All communication is designed to operate in real time with minimal delay to ensure responsive and reliable user feedback.

1.9.3 In-Scope Features

The scope of the Echora project includes real-time environmental awareness enabled through depth sensing, audio-based spatial feedback for navigation assistance, and haptic feedback for collision avoidance and hand guidance. The system also supports object and interaction target detection, such as doors, handles, and buttons, as well as automatic switching between navigation and interaction modes based on user context.

1.9.4 Out-of-Scope Items

The project explicitly excludes the replacement of traditional mobility aids such as white canes or guide dogs. It does not support autonomous navigation or decision-making without user control, nor does it provide medical diagnosis or treatment functionality. Cloud-based processing and

continuous internet dependency are excluded to ensure offline usability. Additionally, the project does not include long-term user training or rehabilitation programs.

1.9.5 Supported Sustainable Development Goals (SDGs)

The Echora project supports several United Nations Sustainable Development Goals. It contributes to SDG 3 by enhancing safety, independence, and quality of life for Blind and Low Vision users. It aligns with SDG 9 through the application of innovative AI and wearable technologies in assistive systems. The project promotes SDG 10 by reducing inequalities through improved access to navigation and interaction in physical environments. Finally, it supports SDG 11 by enabling inclusive and accessible urban mobility without requiring changes to existing infrastructure.

1.10 Objectives

The objectives of the Echora project define the intended outcomes and guide the design and development process. The system aims to enhance environmental awareness by providing continuous real-time perception of surrounding spaces, support safe navigation in indoor and outdoor environments, and enable precise object interaction by guiding the user's hand toward interaction targets. It seeks to integrate multimodal sensor data into a unified perception model and deliver intuitive audio and haptic feedback that is easy to interpret. The project emphasizes minimizing cognitive load through context-aware adaptation, ensuring real-time performance with low latency, and leveraging mobile computing to reduce hardware cost and improve scalability. The system is designed to maintain modularity for future extensions, complement existing mobility aids rather than replace them, and ultimately promote user independence and confidence.

1.11 Challenges

The development of the Echora system involves several technical, ethical, resource-related, and user experience challenges.

1.11.1 Technical Challenges

From a technical perspective, the system must process multimodal sensor data in real time with minimal latency to ensure timely feedback. Sensor reliability remains a concern, as depth sensing and hand tracking accuracy can degrade in varying lighting conditions or cluttered environments. Designing effective audio and haptic feedback requires balancing informativeness with simplicity to avoid overwhelming the user. Wireless communication stability between wearable hardware and the smartphone is critical for responsiveness, and computational constraints on mobile devices may limit the simultaneous execution of multiple AI models.

1.11.2 Ethical Challenges

Ethical challenges include ensuring user privacy, particularly when camera and sensor data are collected from personal and public environments. The use of face recognition or individual identification requires careful consideration of consent and ethical boundaries. There is also a risk of user dependency if over-reliance on the system reduces situational awareness in the event of system failure. Maintaining user autonomy is essential, ensuring that the system assists rather than enforces actions.

1.11.3 Resource-Related Challenges

Resource-related challenges arise from limited development time, which may restrict extensive testing and optimization. Hardware availability can be constrained by cost and supply limitations,

and continuous sensing and processing can significantly impact battery life during prolonged use.

1.11.4 User Experience Challenges

From a user experience perspective, learnability remains a key challenge, as users may require time to adapt to new audio and haptic cues. Cognitive load must be carefully managed to prevent information overload, and wearable components must remain comfortable and unobtrusive to support extended everyday use.

1.12 Risks

Ref	Risk Type	Risk Description	Severity	Likelihood	Impact	Mitigations/Remediations
1	Technical	Real-time processing latency exceeds acceptable limits.	Intolerable	Possible	High	Optimize AI models, reduce resolution,
2	Technical	Depth sensing accuracy degrades in poor lighting or cluttered environments.	Undesirable	Possible	Medium	Optimize AI models, reduce input resolution, prioritize
3	Technical	Wireless communication instability between wearable and smartphone.	Undesirable	Possible	Medium	Use dual links (Wi-Fi a BLE) with buffering.
4	Technical	Battery drains too quickly during continuous operation.	Undesirable	Probable	Medium	Use dual links (Wi-Fi a BLE) with buffering.
5	Hardware	Sensor malfunction or inaccurate readings from depth or sonar sensors.	Undesirable	Possible	Medium	Use dual links (Wi-Fi a BLE) with retry logic.
6	Hardware	Battery drains too quickly during continuous operation.	Undesirable	Probable	Medium	Optimize duty cycles, reduce sensor sampling.
7	Hardware	Sensor malfunction or inaccurate readings from depth or sonar sensors.	Undesirable	Possible	Medium	Optimize duty cycles, reduce sensor sampling and
8	Hardware	Wearable discomfort due to weight or form factor.	Undesirable	Possible	Medium	Offload computation routines, sare,
9	Development	Integration issues between hardware and software components.	Intolerable	Possible	High	Define interfaces early and cycle test
10	Development	Insufficient testing leads to undetected bugs or system failures.	Intolerable	Possible	High	Adopt automated and manual testing.
11	User Experience	Audio or haptic feedback causes cognitive overload.	Undesirable	Possible	Medium	Use saliency prioritization and provide simple onscreen
12	Ethical	Privacy concerns related to camera based sensing	Tolerable	Possible	Medium	Perform core security evaluation.
13	Project	Time constraints limit full implementation or evaluation.	Undesirable	Probable	Medium	Prioritize core features and clear evaluation

Table 1.1: List of Risks

1.13 Standards

Table: List of Standards (Echora)

Category	Standard / Regulation	Description
Software Requirements	ISO/IEC/IEEE 29148	Guidance for writing, managing, and verifying software/system requirements specifications.
Software Life Cycle	ISO/IEC/IEEE 12207	Defines software life-cycle processes from planning through development, operation, and maintenance.
Usability / HCI	ISO 9241-210	Human-centred design principles to improve usability and reduce cognitive load in interactive systems.
Accessibility	EN 301 549	Accessibility requirements for ICT products and services, commonly used for public-sector procurement.
Accessibility (Content)	WCAG 2.2	Web/app accessibility guidelines (perceivable, operable, understandable, robust) for the mobile app UI.
Privacy	GDPR (EU)	Personal data protection rules; relevant for camera-based sensing and any stored user profiles or logs.
Privacy	CCPA/CPRA (California)	Consumer privacy requirements that may apply depending on deployment region and data practices.
AI Risk Management	NIST AI RMF 1.0	Framework to identify, assess, and manage risks of AI systems (bias, safety, transparency, accountability).
AI Risk Management	ISO/IEC 23894	Guidance for AI risk management across the AI system life cycle.
Wireless (Wi-Fi)	IEEE 802.11	Wi-Fi standards supporting high-bandwidth data transfer such as depth frames to the smartphone.
Wireless (Bluetooth)	Bluetooth Low Energy (BLE) Core Spec	Low-power wireless standard suitable for commands, sensor telemetry, and haptic control.
Security	ISO/IEC 27001	Information security management controls to guide secure handling of data and device communication.

Table 1.2: List of Standards

1.14 Constraints

Table 1 – List of Constraints

Constraint	Definition	Description
Technical Constraints	Limitations related to technology, systems, or infrastructure used in the project.	- Hardware limitations (e.g., processing power, battery life), affordability of wearable devices, - limited bandwidth for real-time processing and feedback, - dependency on external APIs and services for data acquisition.
Ethical Constraints	Constraints arising from moral principles and societal values, focusing on the impact on users and stakeholders.	- Ensure user privacy, addressing biases in AI algorithms, and providing clear and understandable feedback to the BLV users that is accurate and safe.
Ethical Constraints		- Ensure compliance with accessibility standards, obtain necessary regulatory approvals for aTIS.
Realistic Constraints	Requirements mandated by laws, regulations, or standards that organizations or individuals must follow.	- Ensure compliance with accessibility standards, and obtain necessary regulatory approvals for assistive technology devices.
Legal Constraints	Requirements mandated by laws, regulations, or standards that organizations or individuals must follow.	- Ensure compliance with data protection laws, such as the GDPR, ensure user data is handled securely and with explicit consent.
Operational Constraints	Rules or limitations that the system's operations must adhere to.	- Weight and comfort limitations for wearable devices, battery life to ensure devices operate throughout the day. - Possible maintenance and calibration requirements to keep the system functioning effectively.

Table 1.3: List of Constraints

1.15 Summary

This chapter introduced the Echora project and established the foundation for the remainder of the document. It outlined the purpose, intended use, and scope of the system, identifying the key challenges faced by Blind and Low Vision users in navigation and object interaction. The chapter also described the project's objectives, system architecture overview, communication approach, and operational constraints.

In addition, potential challenges, risks, applicable standards, and constraints were discussed to highlight technical, ethical, and practical considerations influencing system design and development. Together, these elements define the problem space and clarify the boundaries within which Echora is developed.

The following chapter presents a detailed market analysis and review of related work, situating Echora within the broader context of existing assistive technologies and research efforts.

Chapter 2

Market Analysis and Related Work

2.1 Introduction

This chapter situates the Echora project within the broader context of assistive technologies for Blind and Low Vision (BLV) users. It examines the current market landscape, existing solutions, and relevant research efforts to identify gaps, limitations, and opportunities that motivate the proposed system.

The market analysis explores commercial and emerging assistive products, evaluating their strengths, weaknesses, and positioning through structured analytical frameworks. This analysis provides insight into user needs, technological trends, and external factors influencing the development and adoption of assistive systems.

In addition, this chapter reviews relevant academic literature and related systems that address sensory substitution, navigation assistance, and object interaction. By analyzing prior work and existing approaches, the chapter highlights how Echora differs from and builds upon current solutions.

Together, the market analysis, literature review, and related work establish a comprehensive foundation for understanding the need for Echora and justify its design choices. The findings of this chapter inform the system design and implementation presented in subsequent chapters.

2.2 Market Analysis

The market analysis for Echora examines the current landscape of assistive technologies designed for Blind and Low Vision (BLV) users, with particular emphasis on navigation, object interaction, and sensory substitution systems. Existing solutions range from OCR-based smart glasses and wearable obstacle detectors to smartphone-assisted navigation aids. While these systems address specific user needs, they often focus on a single modality or task, such as text reading or basic obstacle detection.

This analysis evaluates the strengths, limitations, and positioning of current solutions to identify gaps in functionality, usability, and integration. By applying structured analytical frameworks, including SWOT, the 7P's marketing mix, and PESTEL analysis, the section highlights both internal and external factors influencing product adoption and technological feasibility.

The insights gained from this analysis inform the design decisions of Echora and clarify how the proposed system differentiates itself through sensor fusion, multimodal feedback, and support for both macro-navigation and micro-interaction. The following subsections present these analyses and visual comparisons in detail.

2.2.1 SWOT Analysis

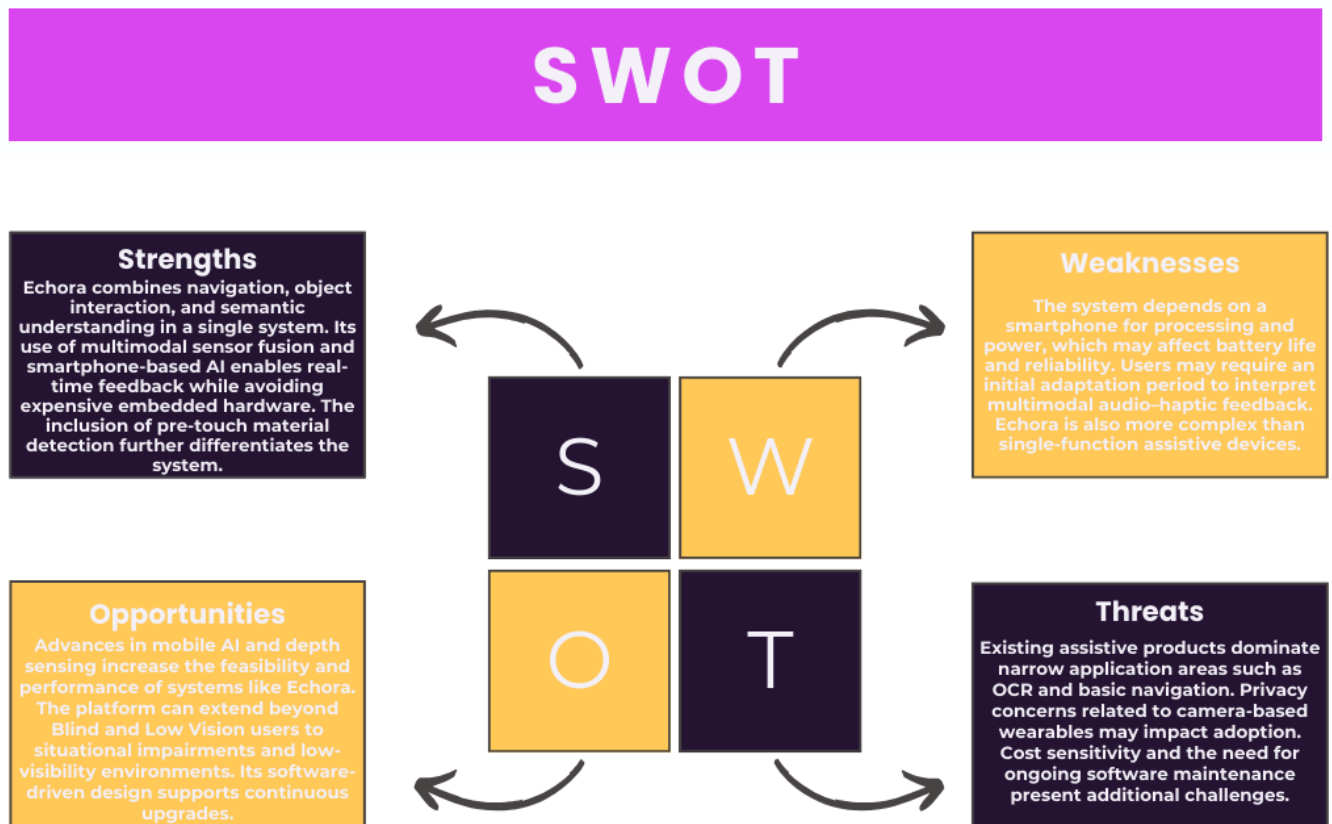


Figure 2.1: SWOT Analysis Diagram

2.2.2 7 P's

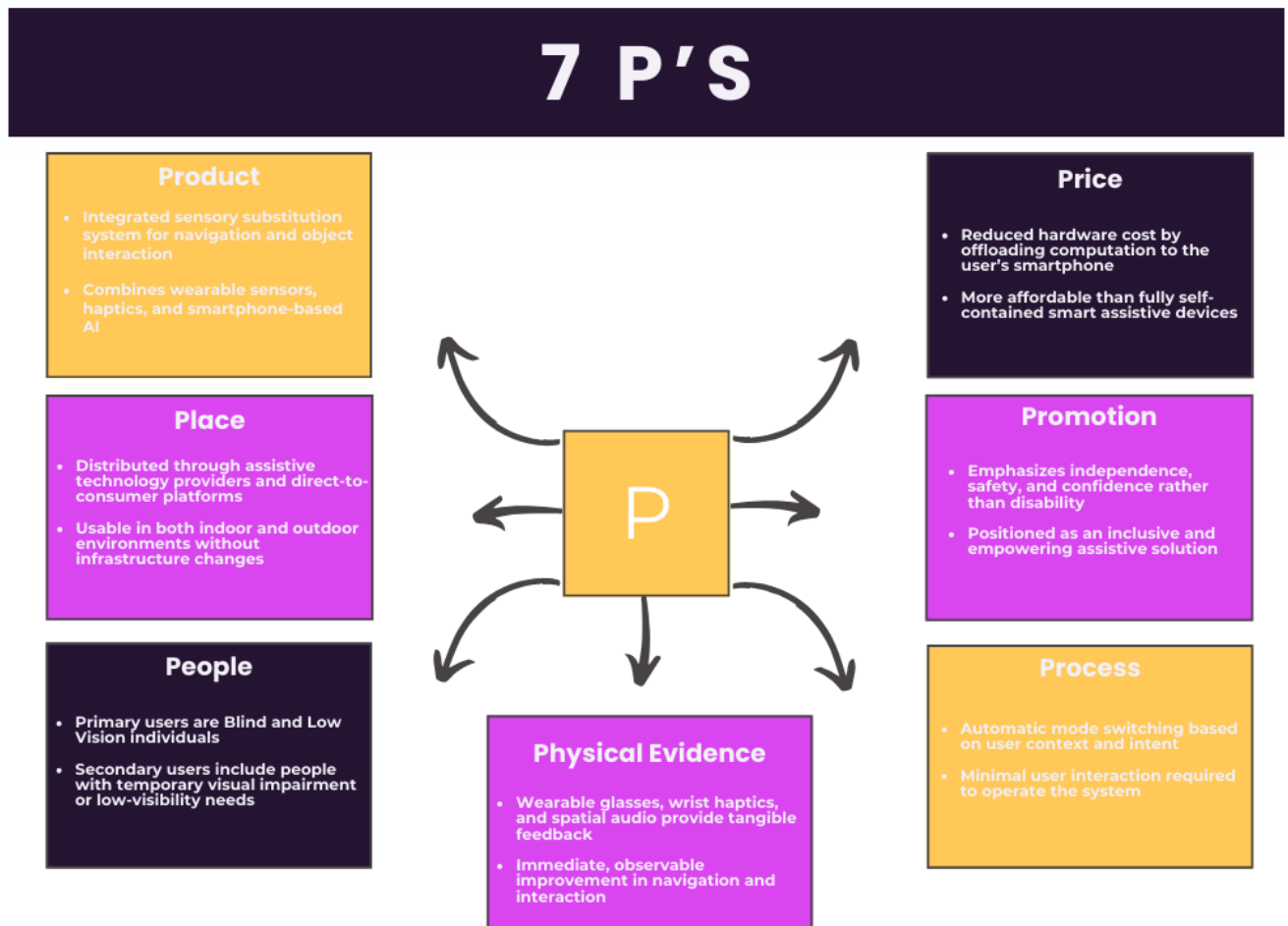


Figure 2.2: 7P's Diagram

2.2.3 PESTEL Analysis

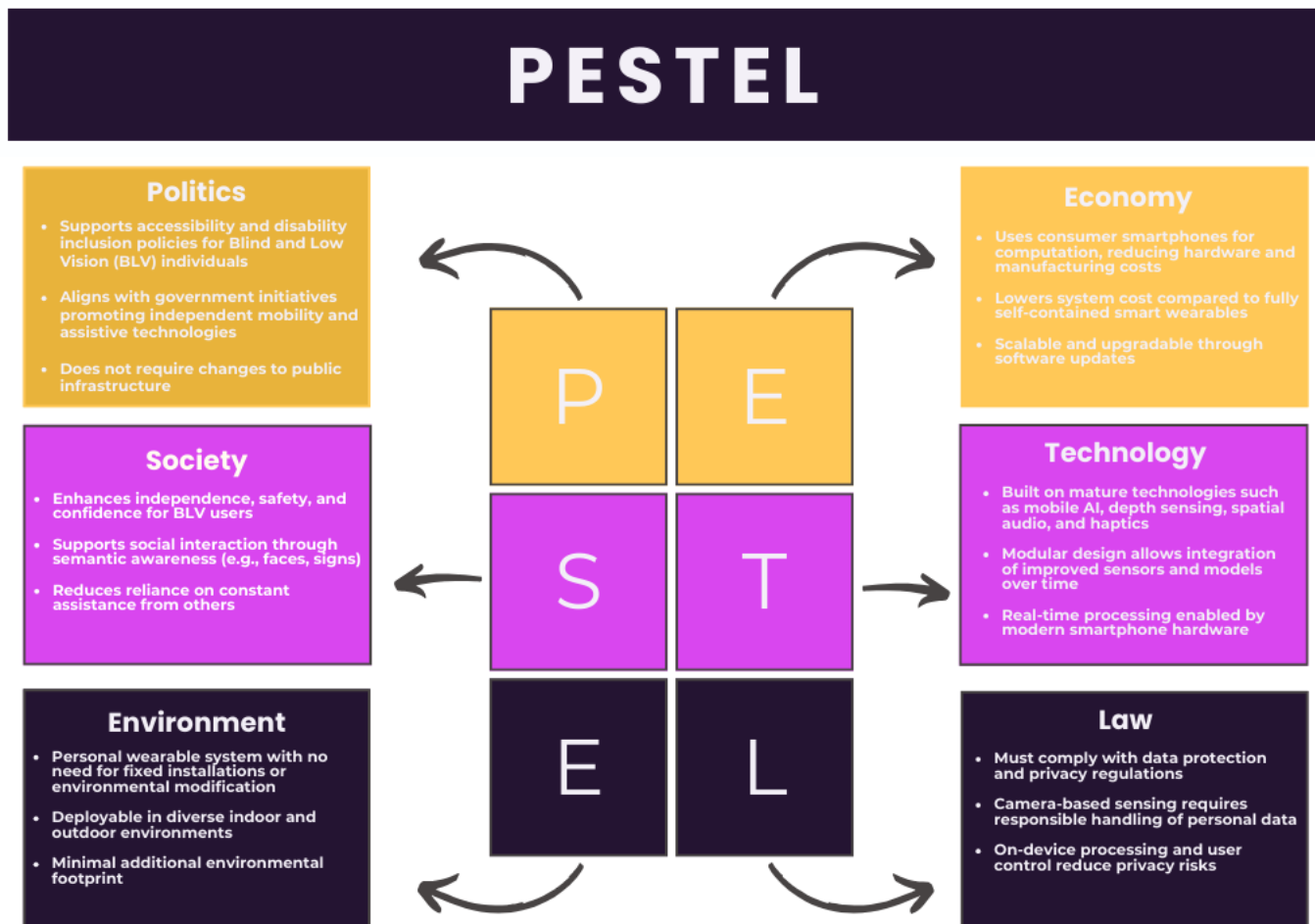


Figure 2.3: PESTEL Analysis Diagram

2.2.4 Similar Projects

Envision Glasses



Figure 2.4: Envision Glasses

Envision Glasses Summary

Description: Envision Glasses are smart glasses designed primarily for text recognition and scene description. The system relies on a camera and smartphone-based AI to provide audio feedback about visible text and objects.

The key features of the system include optical character recognition (OCR), object and scene description, and face recognition. The main advantages of Envision Glasses include strong text-reading performance, a well-established commercial product ecosystem, and ease of use for reading and identification tasks. However, the system does not provide 3D spatial navigation support and does not include haptic feedback. It also does not offer hand guidance or object interaction support, and it lacks depth sensing, which limits its ability to provide continuous spatial awareness.

Seleste Glasses



Figure 2.5: Seleste Glasses

Description: Seleste Glasses are lightweight smart glasses that depend heavily on smartphone vision processing. The system focuses on visual interpretation but offers limited spatial understanding.

The system provides camera-based scene interpretation supported by smartphone AI processing. Its advantages include lightweight hardware design and the flexibility of leveraging smartphone compute. Nevertheless, the system does not provide 3D navigation or depth awareness and does not include haptic feedback. It also lacks object interaction features and hand guidance, resulting in limited assistive functionality when compared to integrated perception systems.

OrCam MyEye



Figure 2.6: OrCam MyEye

OrCam MyEye Summary

Description: OrCam MyEye is a wearable camera that clips onto glasses and provides audio feedback for text reading and face recognition. It is designed for quick access to visual information rather than spatial awareness.

The key functions include OCR for printed and digital text, face recognition, and object identification. The system achieves high-quality OCR performance and is designed to be compact, lightweight, and usable with minimal interaction. However, it does not support navigation or obstacle avoidance, and it does not provide depth perception. Additionally, it does not include haptic feedback or interaction support, and its assistance is largely limited to semantic outputs rather than continuous spatial understanding.

WeWalk Smart Cane



Figure 2.7: WeWalk Smart Cane

WeWalk Smart Cane Summary

Description: WeWalk is a smart cane that enhances traditional white cane use with ultrasonic sensors and vibration feedback to detect nearby obstacles.

The system supports ultrasonic obstacle detection, vibrotactile feedback, and smartphone connectivity for extended functionality. Its main strengths include a familiar form factor for BLV users, simple and intuitive obstacle alerts, and strong compatibility with conventional cane usage. However, its sensing is limited to basic forward obstacle detection and it does not provide semantic understanding of objects. It also does not support hand guidance or object interaction tasks, and it does not construct a 3D environmental map.

Sunu Band



Figure 2.8: Sunu Band

Sunu Band Summary

Description: Sunu Band is a wrist-worn ultrasonic device that alerts users to nearby obstacles using vibration patterns. It is designed for basic spatial awareness rather than detailed perception.

The device provides ultrasonic proximity sensing and vibrotactile alerts in a compact wrist-worn design. Its advantages include portability, low wearability burden, and a simple feedback mechanism that is useful for short-range obstacle detection. Despite these strengths, the device does not provide object recognition or classification and does not perform depth mapping or navigation planning. It also does not support hand guidance or semantic feedback, resulting in limited environmental understanding.

Table 5: Similar Projects Comparison (Including Echora)

Project	Summary	Key Features	Pros	Cons
Envision Glasses	Smart glasses focused on OCR and scene	• OCR • Scene description • Face recognition	• Strong text reading • Mature commercial product	• No navigation • No haptics or hand guidance
Seleste Glasses	Lightweight smart glasses relying on s	• Camera-based interpretation • Mobile AI	• Lightweight • Flexible processing	• No depth • Limited assistance
OrCam MyEye	Wearable camera for instant text and f	• OCR • Face recognition • Object ID	• Excellent OCR • Easy to use	• No navigation • No interaction
WeWalk Smart Cane	Smart cane with ultrasonic obstacle de	• Ultrasonic sensing • Vibration alerts	• Familiar form • Simple alerts	• Basic detection only • No semantics
Sunu Band	Wrist-worn ultrasonic obstacle awarene	• Ultrasonic proximity • Vibrotactile alerts	• Compact • Portable	• No mapping • No object understanding
Echora (Proposed)	AI-powered sensory substitution system	• 3D depth navigation • Haptic hand guidance • Material detection • Sensor fusion	• Navigation + interaction • Pre-touch material awareness • Multimodal feedback	• Prototype stage • Requires smartphone processing

Table 2.1: Comparison of Existing Assistive Technologies and the Proposed Echora System

2.3 Literature Review

This section reviews prior work that informs Echora’s design, focusing on spatial-audio navigation, learnable visual-to-audio mappings, robust 3D world modeling, haptic wearables for guidance, and hand-centered interaction systems. Together, these works motivate Echora’s sensor-fusion approach and the separation between macro-navigation and micro-interaction.

Spatial Audio for Navigation and Obstacle Avoidance

A core direction in assistive navigation is converting spatial structure into 3D audio cues that represent obstacle proximity and layout. EchoSee introduces a mobile navigation approach that aligns closely with Echora’s macro-navigation goal by using an iPhone LiDAR scanner to build a live 3D map and applying raycasting to place virtual audio sources on surrounding surfaces, producing a spatial-audio soundscape. In feasibility testing, EchoSee reported a substantial reduction in obstacle collisions of approximately 40%, indicating that continuous spatial sonification can support safer mobility without relying only on spoken alerts [?]. This supports Echora’s approach of building a continuous sound-world rather than using discrete warning messages alone.

Visual-to-Auditory Sensory Substitution as a Learnable Mapping

Another established strategy maps visual depth and geometry to simple audio parameters that users can learn quickly. Easy-Access SoundView converts a small set of visual points into audio cues where vertical position maps to pitch and depth maps to volume. The study reports that the mapping is easy to learn and that blind participants could navigate obstacle courses from early trials, validating that minimal and intuitive audio encodings can be effective when carefully designed [?]. For Echora, this work provides a practical baseline mapping strategy for early

prototyping before moving to richer 3D generative audio scenes.

Robust Environment Mapping as the Foundation for Assistive World Models

Reliable macro-navigation requires a stable spatial model of the environment. RGB-D mapping approaches build dense 3D indoor maps using depth alignment together with visual feature tracking to maintain accuracy even in challenging conditions such as low-texture surfaces or poor lighting. This directly supports Echora's use of depth and vision fusion to improve map stability, which in turn affects the quality of audio placement and collision warnings in a sound-world navigation system [?].

Haptic Wearables for Sensory Replacement and Guidance

Haptic feedback is widely used to reduce cognitive load from audio and to provide directional or proximity cues through the body. A broad review of haptic wearables categorizes systems into sensory replacement, sensory augmentation, and training devices, and surveys multiple applications including navigation and obstacle avoidance. These findings support Echora's wrist-based haptic approach, in which a wrist-mounted haptic actuator is used to deliver directional and proximity feedback for navigation, collision avoidance, and fine-grained task-level guidance [?].

Hands-Based Tactile Perception for Object Interaction (Micro-Interaction)

While many assistive systems focus primarily on navigation, Seeing With the Hands (CHI '25) shifts sensory substitution toward manual interaction, arguing that the hand's perspective is ergonomic and flexible for object exploration. The authors prototype a wrist-mounted camera and render tactile images through an electrotactile array on the back of the hand, enabling Blind and Low Vision users to complete object-related tasks. Importantly, participants preferred combining both eye-view and hand-view perspectives, using one for overview and the other for detail. This directly motivates Echora's separation of macro-navigation and micro-interaction feedback, and supports multimodal fusion rather than replacing one modality entirely [?].

Smart Glasses Prototypes for OCR-Based Assistance and Practical Constraints

OCR-centered smart-glasses prototypes demonstrate the value of text access but also reveal practical constraints relevant to Echora. A smart-glasses system using text detection and OCR reports strong text recognition performance, but also highlights real-world limitations such as language support constraints, effective capture distance, and ergonomic considerations for mounting and comfort. For Echora, these findings emphasize the importance of wearable comfort, robust sensing under varied conditions, and fallback strategies when camera-based perception is unreliable [?].

Deep Learning Wearables for Navigation (Vision + Sensors)

Recent wearable navigation systems show that real-time deep learning can be feasible when models are lightweight and when sensors are combined. A low-cost wearable and smart-cane hybrid integrates vision-based perception with ultrasonic sensing and GPS augmentation, and reports comparative results indicating that modern object-detection approaches can achieve practical accuracy and frame rates. For Echora, this reinforces the feasibility of on-device real-time inference and supports the project's emphasis on sensor fusion to improve robustness and safety compared to relying on a single sensing modality [?].

Literature Review Summary Table

Title, Author and Year	Objective	Method	Result	Dataset	Limitations
EchoSee: Spatial Audio Navigation Using Mobile LIDAR , et al., 2022	Assist BLV users in navigation by converting 3D spatial structure into continuous spatial audio cues.	Uses iPhone LIDAR to build a real-time 3D map and applies raycasting to position virtual spatial audio sources on surround.	Significant reduction in obstacle collisions (~40%); demonstrated effectiveness of continuous spatial sonification.	Real-world indoor environments, captured using mobile LIDAR.	Requires LIDAR-equipped devices; limited evaluation in complex outdoor environments.
Easy-Access SoundView , et al., 2018	Provide a simple and learnable visual-to-audio mapping for obstacle avoidance by BLV users.	Maps visual depth and vertical position to audio parameters such as volume and pitch using a small set of sampled points.	Users quickly learned the mapping and successfully navigated obstacle courses from early trials.	Real-world indoor environments.	Controlled experimental environments.
RGB-D Mapping for Robust Indoor Environment Modeling , Kumar & Jain, 2016	Build stable and accurate 3D environment maps for navigation and interaction.	Combines depth alignment with visual feature tracking to maintain robust RGB-D maps under challenging conditions.	Improved mapping stability in low-texture and low-light environments.	RGB-D indoor datasets.	High computational cost; primarily designed for indoor settings.
Haptic Wearables for Sensory Replacement and Guidance , Shull & Damian, 2015	Survey and classify haptic wearable systems for navigation guidance and sensory substitution.		High computational cost; primarily designed for indoor settings.	RGB-D indoor datasets.	Review of multiple prior studies.
Seeing With the Hands , CHI 2025	Enable BLV users to interact with objects using hand-centered tactile perception.	Comprehensive review of haptic feedback modalities and body location for conveying directional.	Identified haptics as effective for reducing audio load and improving directional guidance.	Review of multiple prior studies.	No unified design standard and effectiveness varies by body placement and
Smart Glasses for Blind People Using OCR , et al., 2020	Provide access to printed text for BLV users through smart glasses.	Combines deep-learning-based object detection with OCR and audio output on wearable glasses.	Printed text datasets and real-world testing.	Custom experimental datasets and user study.	Limited field-of-view; not designed for large scale navigation.
Deep Learning-Based Wearable Navigation System , Kumar & Jain, 2022	Enable real-time navigation assistance using wearable vision and sensors.	Integrates deep-learning-based object detection with ultrasonic sensors.	Public object-detection datasets and real-world trials.	Printed text datasets and real-world testing.	Sensitive to lighting and capture distance; ergonomic constraints.

Table 2.2: Literature Review Summary Table

2.4 Background

The integration of artificial intelligence (AI), machine learning (ML), and wearable sensing technologies has significantly advanced assistive systems for Blind and Low Vision (BLV) individuals. Modern assistive devices increasingly rely on computer vision, depth sensing, haptics, and audio feedback to compensate for the loss of visual perception. These technologies aim to provide users with spatial awareness, object recognition, and navigation support in real time, enabling safer and more independent interaction with their environment. However, designing systems that are both computationally efficient and perceptually intuitive remains a key challenge, particularly in wearable and mobile contexts.

Sensory substitution systems translate visual information into alternative sensory modalities such as sound or touch. Early sensory substitution approaches demonstrated that spatial layouts and object features could be conveyed through auditory or tactile feedback. Recent advances have expanded these systems using depth cameras, spatial audio, and haptic interfaces to improve environmental understanding. Despite their effectiveness in navigation tasks, many sensory substitution systems focus primarily on global spatial awareness and provide limited support for fine-grained object interaction.

Computer vision and depth sensing play a central role in modern assistive technologies. Depth cameras and RGB-D sensors enable the reconstruction of three-dimensional representations of the environment, supporting obstacle detection, distance estimation, and spatial mapping. These techniques allow systems to perceive structural elements such as walls, doors, and furniture. However, processing depth data in real time requires efficient algorithms and hardware acceleration, especially when deployed on mobile devices with limited computational resources.

Machine learning models, particularly Convolutional Neural Networks (CNNs), are widely used for visual perception tasks such as object detection, text recognition, and scene understanding. CNNs excel at extracting spatial features from images and depth maps, making them suitable for assistive applications that require real-time perception. While these models provide high accuracy, their computational demands pose challenges for wearable systems, motivating the use of smartphone-based processing rather than fully embedded solutions.

Haptic feedback and hand guidance have been explored as effective methods for conveying spatial and directional information non-visually. Vibrotactile and electrotactile interfaces can guide user motion by encoding direction, distance, and urgency through tactile patterns. Hand-centered haptic guidance has been shown to support precise object interaction by providing feedback relative to the user's hand position. However, most existing systems rely on predefined targets or simple proximity cues, limiting their flexibility in unstructured environments.

Multimodal sensor fusion combines data from vision, depth, inertial sensors, and active acoustics to provide a more robust and comprehensive perception of the environment. By integrating multiple sensory inputs, assistive systems can adapt feedback based on user context and intent. Nevertheless, achieving low-latency fusion and intuitive feedback remains challenging, particularly in real-time mobile applications.

In summary, prior research highlights significant progress in sensory substitution, computer vision, haptics, and machine learning for assistive technologies. However, challenges related to real-time performance, fine-grained interaction, and pre-contact object understanding persist. These limitations motivate the design of integrated systems, such as Echora, that unify navigation, interaction, and semantic understanding through a single adaptive framework.

2.5 Summary

This chapter examined the market landscape, existing assistive technologies, and relevant research related to sensory substitution systems for Blind and Low Vision users. The market analysis showed that current commercial solutions are largely fragmented, with most devices addressing isolated tasks such as text recognition or basic obstacle detection. SWOT, PESTEL, and marketing analyses highlighted both the opportunities enabled by advances in mobile AI and wearable sensing, as well as the limitations and challenges faced by existing systems.

The literature review and related work demonstrated that sensory substitution, depth-based perception, haptic guidance, and mobile-based AI are viable approaches for supporting navigation and interaction. However, prior systems typically focus on either environmental awareness or object-level interaction, rarely combining both within a single adaptive framework.

These findings motivate the design of Echora as an integrated sensory substitution system that unifies navigation, hand-level interaction, and semantic understanding through real-time multimodal sensor fusion. The next chapter presents the system design and architecture of Echora, detailing how these concepts are realized in practice.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the methodology and system analysis adopted for the design and development of Echora, an AI-powered sensory substitution system for Blind and Low Vision (BLV) users. It describes the structured approach used to transform project objectives and user needs into a coherent technical solution, ensuring feasibility, usability, and alignment with real-world constraints.

The chapter outlines the overall development methodology, including the rationale behind key design decisions, the validation of project relevance through proof of interest, and the planning strategies employed to manage complexity and risk. Emphasis is placed on iterative development, user-centered design principles, and continuous evaluation to ensure that the system effectively supports both macro-navigation and micro-interaction tasks.

In addition, this chapter details the project planning process, including the project management plan and the adoption of Agile practices. These practices enable flexibility, incremental progress, and early integration of feedback, which are critical for developing a real-time, safety-critical assistive system such as Echora. Together, the methodology and planning framework described in this chapter provide a solid foundation for the system design, implementation, and evaluation discussed in subsequent sections.

3.2 Methodology

The development of the Echora system follows a user-centered, iterative methodology that combines principles from systems engineering, human-centered design, and Agile software development. This methodology is selected to address the technical complexity, real-time constraints, and safety-critical nature of assistive technology for Blind and Low Vision (BLV) users.

At the core of this methodology is the transformation of user needs into functional system components through continuous validation and refinement. Rather than treating navigation, object interaction, and perception as isolated problems, Echora is developed as an integrated sensory substitution system in which sensing, processing, and feedback are tightly coupled.

The methodology begins with requirements analysis grounded in user-centered design. The development process starts by analyzing user needs, accessibility requirements, and limitations of existing assistive technologies. Functional and non-functional requirements are derived from the challenges faced by BLV users in navigation and object interaction, from safety, usability, and accessibility considerations, and from insights obtained through related research and market analysis. These requirements guide subsequent design and implementation decisions, ensuring

that system development remains focused on real-world usability rather than purely technical performance.

Following requirement definition, Echora is decomposed into independent yet interconnected modules to support modular system engineering. The system is structured around sensor acquisition and synchronization across depth, RGB, and inertial inputs, perception and AI processing for mapping, detection, and recognition, context-awareness and mode switching between navigation and interaction, and feedback generation through spatial audio and haptics. This modular architecture enables components to be developed, tested, and optimized independently while maintaining a clear pathway toward scalable system integration.

After modularization, development proceeds through iterative prototyping and incremental implementation. The system is built through successive prototypes, where each iteration introduces a limited set of features—such as navigation audio, hand guidance, or text recognition—that are tested individually before integration into the full system. This approach enables early identification of performance bottlenecks, usability concerns, and sensor limitations. Feedback from testing and evaluation is used to refine system behavior, improve feedback strategies, and optimize real-time performance.

A core methodological principle in Echora is multimodal sensor fusion combined with real-time on-device processing. Depth, vision, inertial data, and other sensing signals are integrated to construct a unified perception of the environment. Processing is performed entirely on-device to satisfy latency constraints, protect user privacy, and improve system reliability in environments without stable connectivity. Real-time constraints are treated as an explicit design requirement throughout development, and latency, frame rate, and resource usage are monitored continuously to ensure responsive and safe operation.

The methodology also emphasizes feedback design and cognitive load management. Audio and haptic feedback are designed to be intuitive and non-overlapping, with information prioritized by urgency and context. Critical safety alerts are designed to override non-essential feedback when necessary. In addition, automatic mode switching reduces cognitive load by activating only the system functions relevant to the user's current context, enabling smoother transitions between macro-navigation and micro-interaction.

Validation and risk mitigation are integrated across the entire development lifecycle rather than being deferred to a final evaluation stage. Individual modules are validated independently through unit-level tests, followed by system-level integration testing to confirm correct end-to-end behavior. Potential risks, including sensor failure, latency spikes, or feedback overload, are addressed through fallback behaviors and graceful degradation strategies that maintain safety and usability even under degraded conditions.

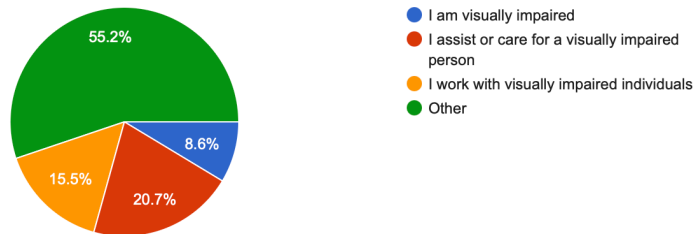
Through this structured yet flexible methodology, Echora is developed as a reliable, adaptive, and user-focused assistive system capable of operating effectively in complex real-world environments.

3.2.1 Proof of Interest

What is your relationship to visual impairment?

[Copy chart](#)

58 responses



How difficult is indoor navigation for visually impaired individuals in unfamiliar environments (e.g., malls, universities, hospitals)?

[Copy chart](#)

58 responses

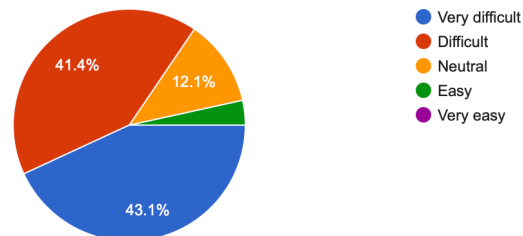
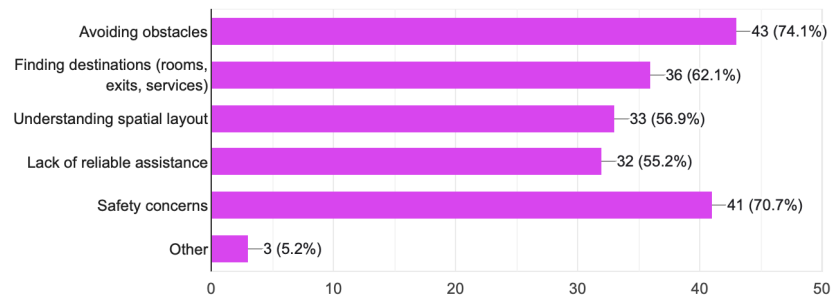


Figure 3.1: Questionnaire Statistics Graphs 1 & 2

What are the most common challenges faced during indoor navigation? *(Select all that apply)*

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58 responses



Which tools are currently used for navigation assistance? *(Select all that apply)*

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58 responses

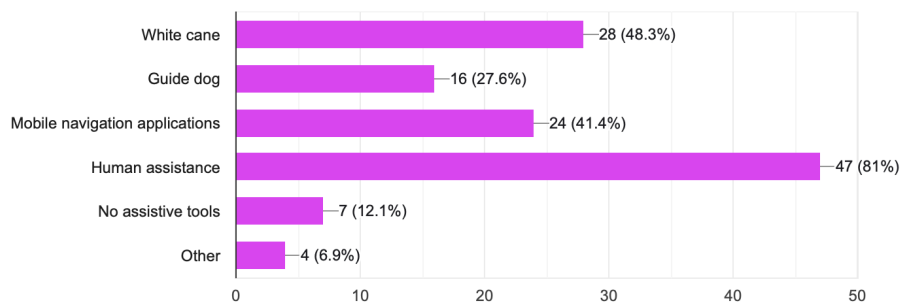
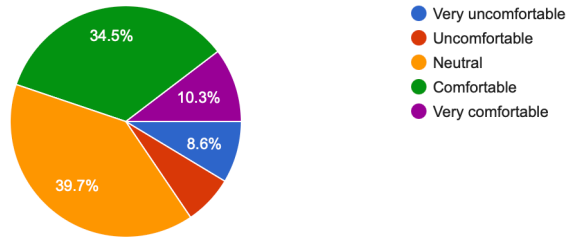


Figure 3.2: Questionnaire Statistics Graphs 3 & 4

How comfortable are you with using smartphone-based assistive technologies?

[Copy chart](#)

58 responses



How helpful do you think audio guidance (e.g., via wireless earphones) would be for indoor navigation?

[Copy chart](#)

58 responses

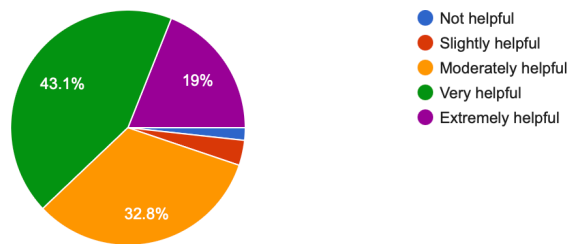
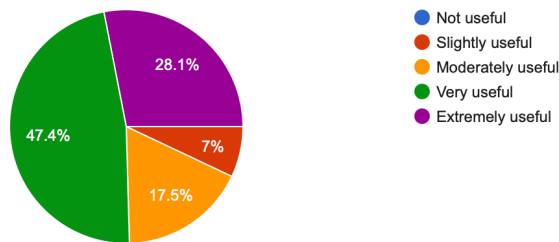


Figure 3.3: Questionnaire Statistics Graphs 5 & 6

How useful would an AI-based indoor navigation system like ECHORA be in daily life?

[Copy chart](#)

57 responses



Which features would be most important in such a system? (Select all that apply)

[Copy chart](#)

58 responses

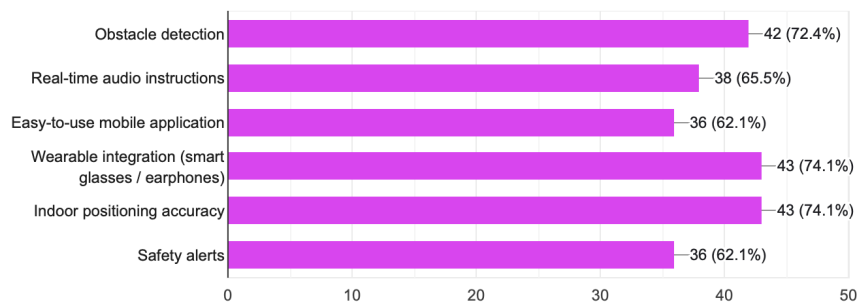
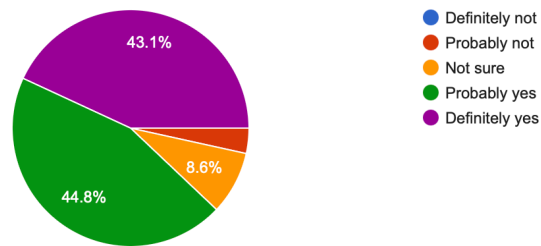


Figure 3.4: Questionnaire Statistics Graphs 7 & 8

Would you be willing to use or recommend a system like ECHORA if it were available?

[Copy chart](#)

58 responses



Overall, how interested are you in using or supporting assistive navigation systems based on artificial intelligence?

[Copy chart](#)

58 responses

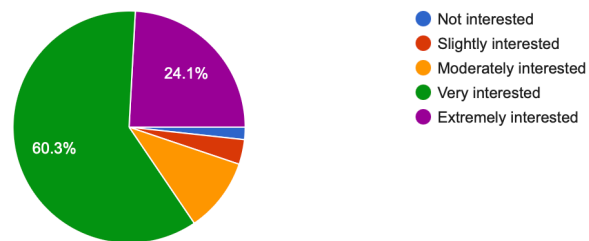


Figure 3.5: Questionnaire Statistics Graphs 9 & 10

3.3 Project Planning

3.3.1 Project Management Plan

Milestone ID	Milestone Name	Planned Completion Date
1	Project Initiation and Planning	2025-10-30
2	Documentation Review and Gap Analysis	2025-11-20
3	Requirements Finalization and System Design	2025-12-22
4	Development: Hardware Prototyping	2026-1-20
5	Ai Training models	2026-2-6
6	Development: Mobile Application Implementation	2026-2-27
7	System Integration	2026-3-20
8	Initial System Testing	2026-4-6
9	Validation and Performance Evaluation	2026-4-22
10	System Refinements and Improvements	2026-5-6
11	Documentation Finalization and Report Preparation	2026-5-17
12	Final Submission and Presentation	2026-5-28

Table 3.1: Project Milestone Table

Phase	Start Date	End Date	Duration (Days)
Project Initiation and Planning	2025-10-1	2025-10-30	30
Documentation Review and Gap Analysis	2025-10-31	2025-11-20	20
Requirements Finalization and System Design	2025-11-21	2025-12-22	31
Development: Hardware Prototyping	2026-1-1	2026-1-20	20
Ai Training models	2026-1-21	2026-2-6	15
Development: Mobile Application Implementation	2026-2-7	2026-2-27	20
System Integration	2026-2-28	2026-3-20	20
Initial System Testing	2026-3-21	2026-4-6	15
Validation and Performance Evaluation	2026-4-7	2026-4-22	15
System Refinements and Improvements	2026-4-21	2026-5-6	15
Documentation Finalization and Report Preparation	2026-5-7	2026-5-17	10
Final Submission and Presentation	2026-5-18	2026-5-28	10

Table 3.2: Project Timeline Table

3.3.2 Agile Project Planning

The Echora project adopts Agile practices to support incremental development, early validation, and continuous adaptation to technical and usability feedback. Agile planning is used to break down complex system requirements into manageable tasks, enabling iterative delivery of system components and improved visibility of progress across documentation and implementation phases. Sprint-based execution supports frequent integration and testing, reducing the likelihood of late-stage integration failures and enabling earlier identification of performance and feasibility risks.

The Agile workflow is managed using a backlog-driven process in which tasks are prioritized based on implementation dependencies, risk level, and expected user impact. The documentation phase backlog is used to organize requirement definition, system specification, and design deliverables, while the implementation phase backlog is used to track development, integration, testing, and optimization tasks. Sprint planning and reviews provide a structured mechanism for assessing progress, updating priorities, and integrating lessons learned into subsequent iterations.



Figure 3.6: Documentation Phase Backlog

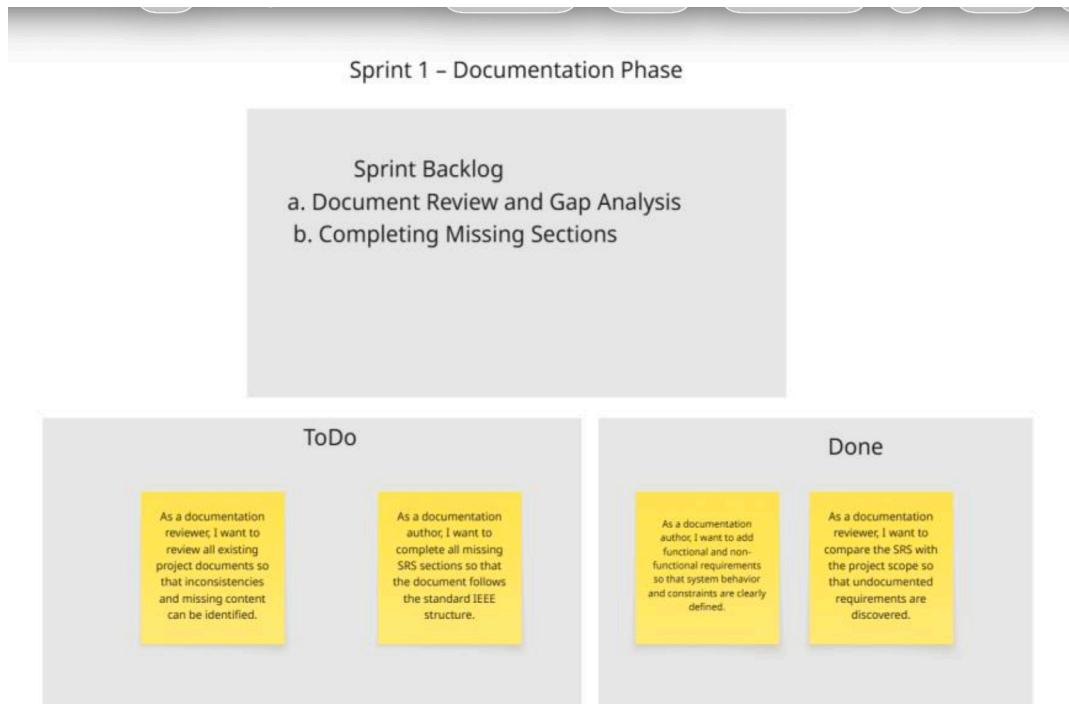


Figure 3.7: Sprint 1 Documentation Phase



Figure 3.8: Sprint 2 Documentation Phase

Sprint 3 – Documentation Phase

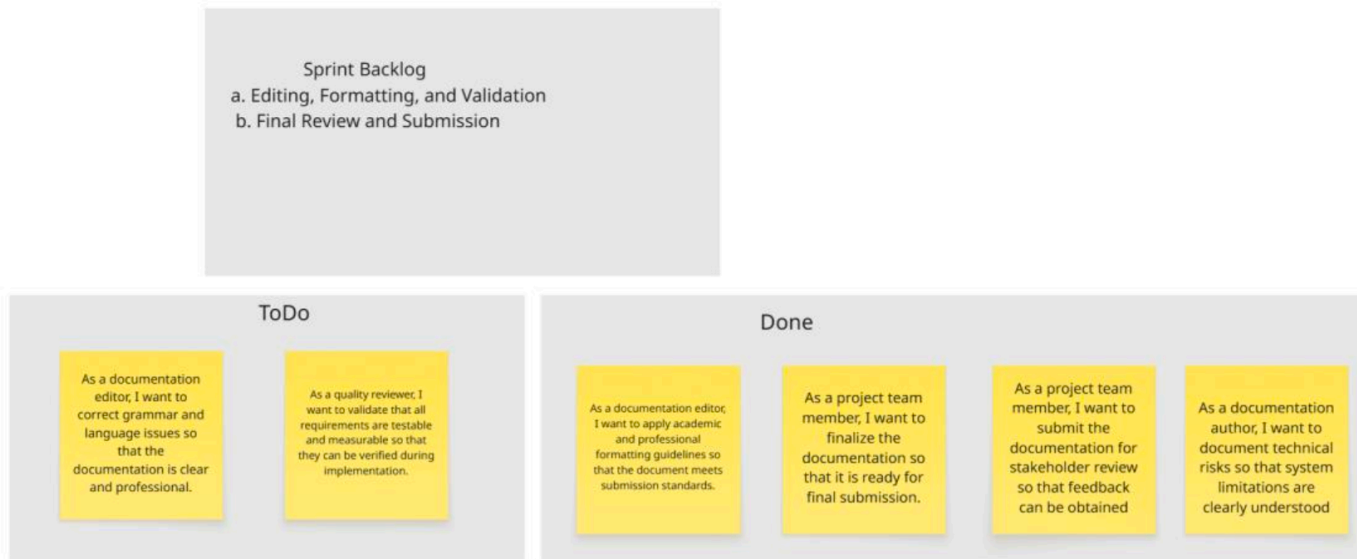


Figure 3.9: Sprint 3 Documentation Phase

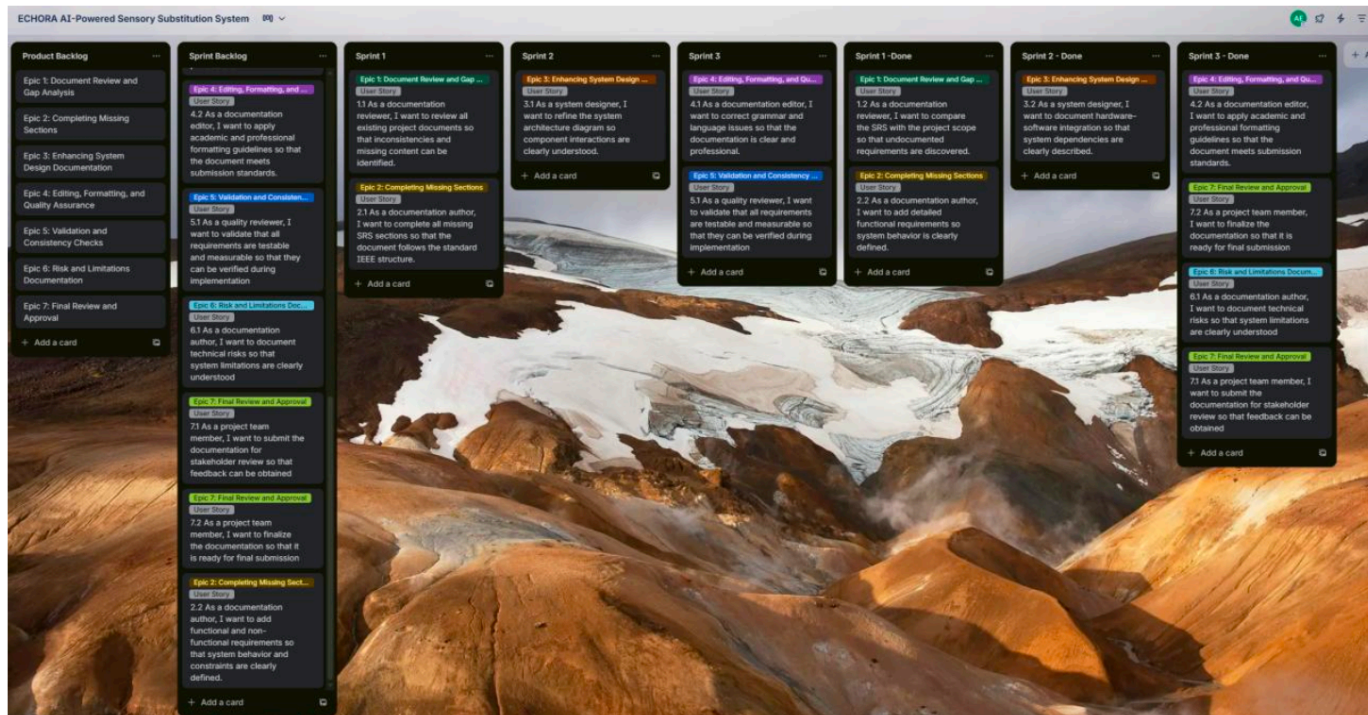


Figure 3.10: Trello Backlog Documentation Phase



Figure 3.11: Implementation Phase Backlog

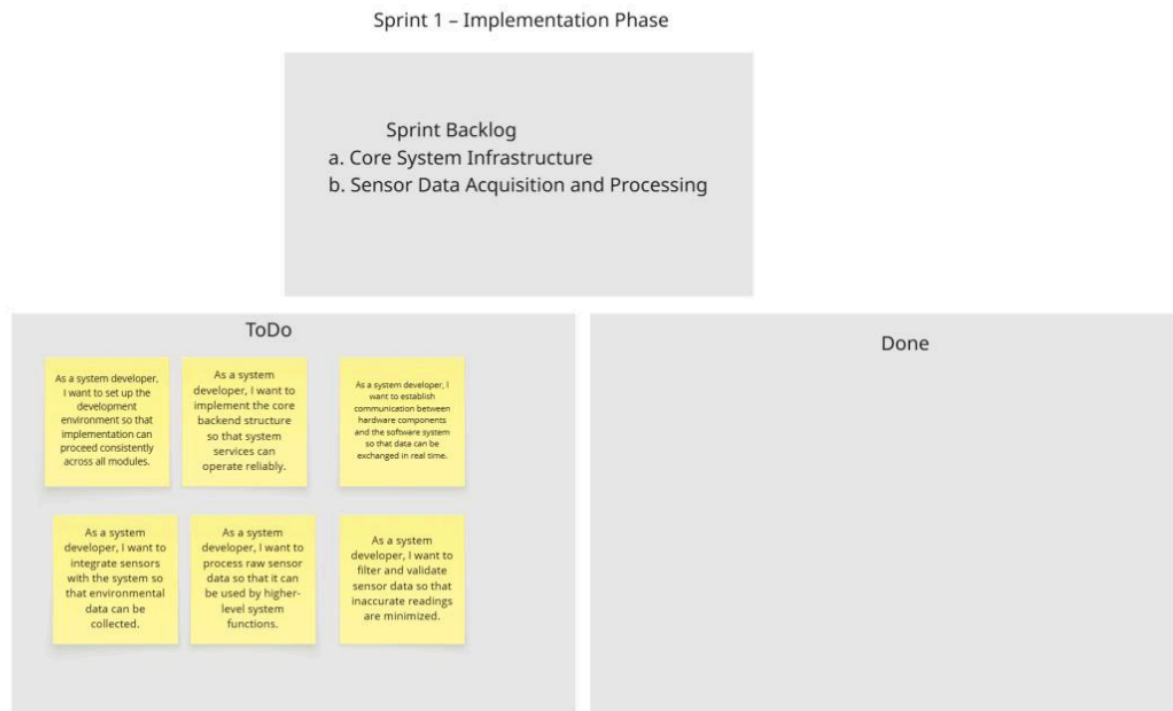


Figure 3.12: Sprint 1 Implementation Phase

Sprint 2 – Implementation Phase



Figure 3.13: Sprint 2 Implementation Phase

Sprint 3 – Implementation Phase

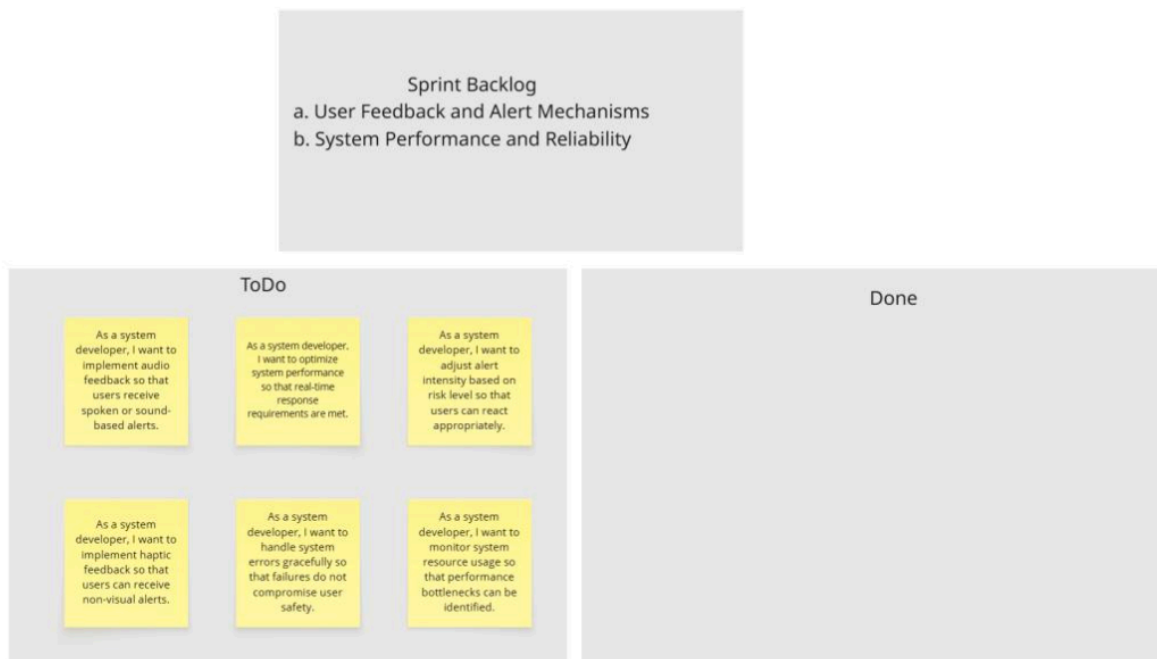


Figure 3.14: Sprint 3 Implementation Phase

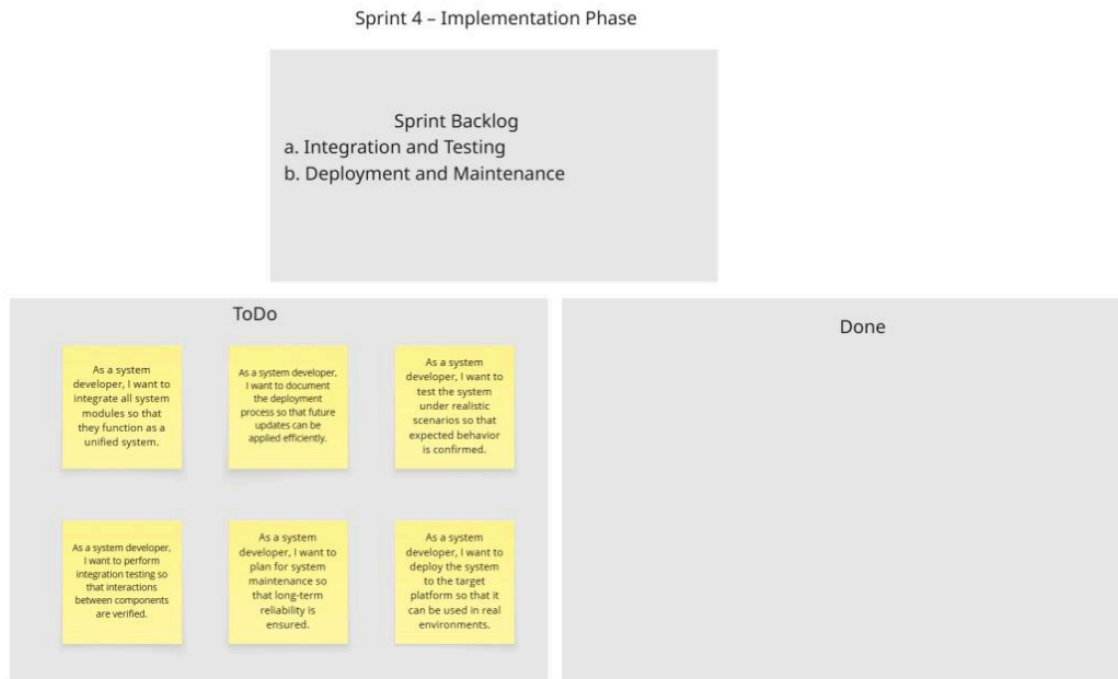


Figure 3.15: Sprint 4 Implementation Phase

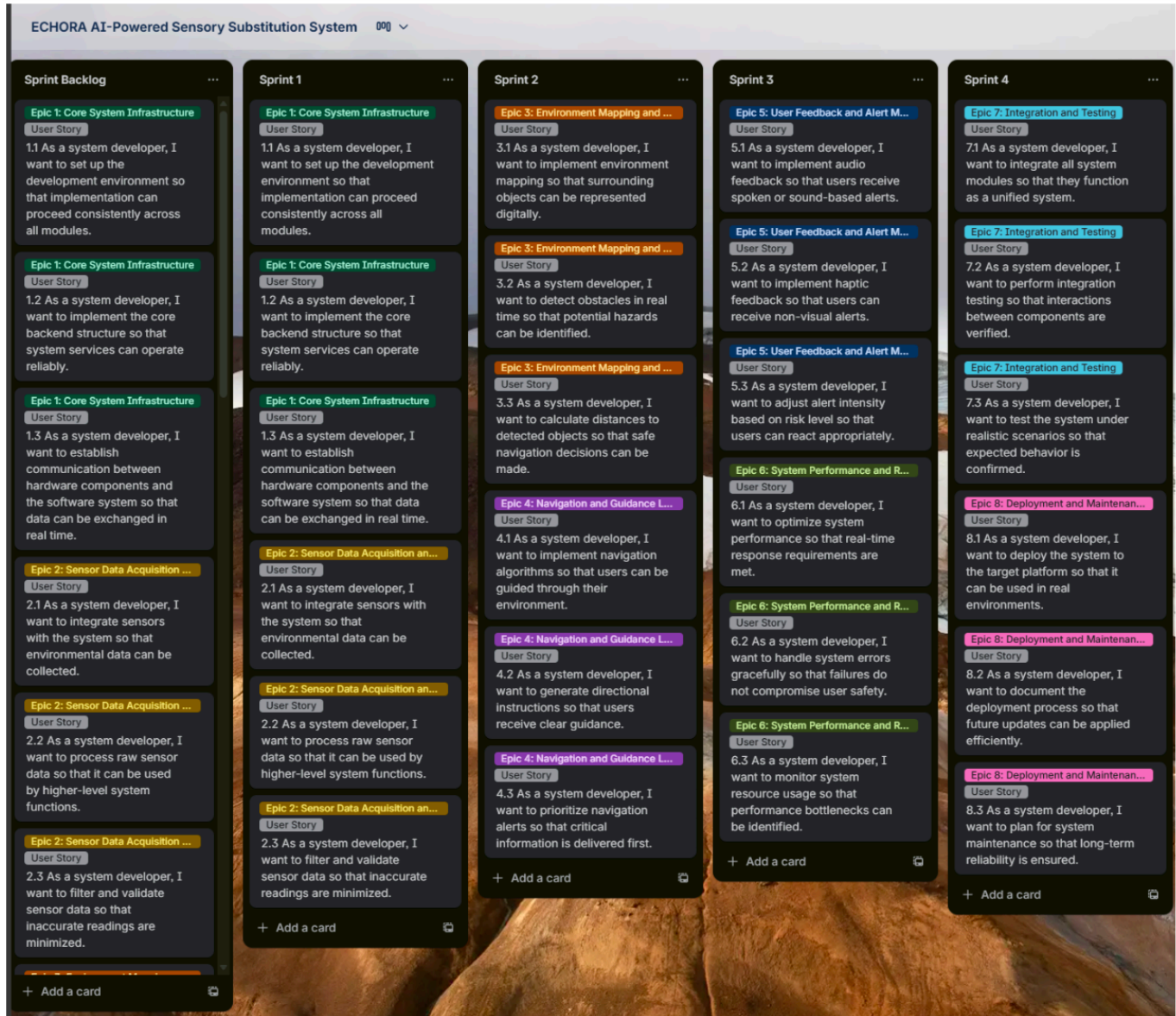


Figure 3.16: Trello Backlog Implementation Phase

Chapter 4

Requirements and System Analysis

4.1 System Features & Functional Requirements

4.1.1 Feature 1: Continuous Environment Mapping & Audio Guidance

Description & Priority

This feature provides continuous, real-time three-dimensional spatial awareness of the surrounding environment, including nearby walls, furniture, and open pathways. It operates by combining depth sensor data with inertial measurements to construct a stable representation of the user's immediate surroundings. The feature enables safe and intuitive walking by generating spatialized audio cues that communicate the direction, distance, and proximity of obstacles and navigable space. Due to its fundamental role in user safety and navigation, this feature is classified as a Critical (Live) priority.

Stimulus	Response
The user is walking in a corridor or open space.	The smartphone processes incoming depth frames and IMU data to generate spatial 3D audio cues whose pitch, tone, and position dynamically change based on the distance and location of nearby obstacles and open areas.

Table 4.1: Stimulus-Response for Continuous Environment Mapping

Functional Requirements

The system shall continuously acquire and fuse sensor data by streaming low-resolution depth frames and high-frequency IMU data from the smart glasses to the smartphone using Wi-Fi Direct. All incoming sensor data shall be timestamped and buffered to maintain synchronized fusion between depth and inertial streams. A sensor fusion algorithm, such as a Kalman Filter, shall be applied to combine IMU signals with depth frames to achieve stable real-time pose and orientation estimation. If sensor data latency exceeds 150 milliseconds, the system shall temporarily switch to IMU-only navigation until depth data resumes. In the event that more than five consecutive depth frames are lost, the system shall attempt automatic reconnection without interrupting user operation.

The smartphone shall perform real-time three-dimensional environment mapping using a lightweight vision model capable of segmenting walls, obstacles, and open pathways. The system shall compute obstacle distance, direction, and relative motion with respect to the user

and classify detected elements into predefined categories including walls, furniture, humans, and open space. Mapping updates shall occur at a minimum rate of 15 frames per second to maintain smooth spatial awareness. If segmentation confidence drops below a defined threshold, the system shall gracefully degrade by switching to coarse shape detection only.

To ensure real-time performance, the system shall downsample distant depth data beyond three meters to prioritize computation near the user while preserving full-resolution processing within the one-to-three-meter safety range. Processing time per frame shall not exceed 40 milliseconds to prevent perceptible audio delays. The system shall dynamically adjust computational load based on CPU and GPU temperature and fall back to simplified segmentation pipelines during performance spikes.

Spatial audio feedback shall be generated to indicate the direction and distance of walls, furniture, and open spaces. Audio cues shall adapt pitch, tone, and intensity based on proximity, with critical obstacles within 1.5 meters triggering stronger and more urgent signals. Audio output shall be managed using a priority-based queue to prevent overlapping cues, and newly detected hazards shall immediately interrupt ongoing non-critical audio.

The navigation feature shall activate automatically when the user is detected to be walking continuously for more than two seconds. Manual activation shall also be supported through a voice command such as “Guide me,” with an audible confirmation stating “Navigation active.” When the user enters a safe, open area, audio intensity shall be reduced to prevent fatigue. Guidance shall pause automatically when the user stops moving and resume once motion is detected again.

From a safety perspective, if depth data is lost for more than 200 milliseconds, the system shall trigger a strong wrist-based haptic alert and switch to IMU-only navigation while announcing “Depth unavailable. Switching to fallback navigation.” If IMU data becomes unstable, navigation guidance shall be halted entirely and the user shall be prompted to adjust the glasses. All error notifications shall be delivered without blocking critical safety cues, and error events shall be logged for debugging and system optimization.

Environment mapping shall operate at a minimum of 15 frames per second and preferably above 24 frames per second. Total end-to-end latency from depth acquisition to audio output shall not exceed 120 milliseconds. All processing shall occur fully on-device without reliance on cloud services, and power consumption shall remain within defined limits to preserve battery life.

4.1.2 Feature 2: Text and Sign Detection

Description & Priority

This feature enables the detection and recognition of text and signs present in the environment, including printed or handwritten text as well as traffic signs, warning signs, icons, and logos. It integrates optical character recognition with object detection and semantic classification to identify text regions, extract their content, and interpret sign meaning. The feature operates entirely on-device and is assigned a High priority due to its importance in information access rather than immediate safety.

Stimulus	Response
A region containing text is detected.	The system extracts the text using OCR and reads it aloud using text-to-speech.
A traffic or warning sign is detected.	The system classifies the sign and verbally announces its meaning to the user.

Table 4.2: Stimulus-Response for Text and Sign Detection

Functional Requirements

The smart glasses shall stream camera frames to the smartphone continuously, and the system shall handle temporary frame loss by retrying or using the most recent valid frame. A real-time text detection model shall identify candidate text regions, filtering out non-text elements based on confidence thresholds. If no text is detected within one second, the system shall announce “No text found” without blocking subsequent detection attempts.

Detected text regions shall be processed using an on-device OCR engine, with preprocessing steps applied to handle noise, skew, and low-light conditions. OCR results below a defined confidence threshold shall be discarded, and the system shall request new frames. Recognized text shall be converted into speech using an on-device text-to-speech engine, with audio playback managed through a non-overlapping queue. If new text appears during playback, the system shall interrupt the current output and read the new content.

The feature shall activate automatically when the user holds an object or sign within approximately 70 centimeters of the camera, and it shall also support manual activation via voice command such as “Read this.” Detection initiation shall be confirmed with an audible “Scanning...” prompt. Error conditions such as camera unavailability, insufficient lighting, or incomplete text visibility shall trigger appropriate user feedback messages.

OCR processing shall operate at a minimum of 10 frames per second, with an ideal target above 20 frames per second. Total latency from detection to spoken output shall not exceed one second.

4.2 System Requirements

This section defines the system-level requirements for Project Echora, describing the functional and operational capabilities required to satisfy user needs. The system architecture consists of wearable smart glasses, haptic feedback devices, and a smartphone-based processing unit.

The wearable system shall consist of smart glasses that are comfortable for extended daily use and integrate depth sensing, RGB imaging, and inertial measurement capabilities. Sensor data shall be streamed continuously to a paired smartphone using secure wireless communication. The system shall include wearable haptic actuators, such as a wrist-mounted device, to deliver directional tactile feedback.

From a perception standpoint, the system shall perform real-time sensor fusion using depth, RGB, and IMU data to generate a continuous three-dimensional representation of the environment. It shall detect obstacles, open pathways, and environmental hazards within a safe navigation range while analyzing proximity, direction, and relative motion.

Navigation assistance shall be delivered using continuous spatial audio cues that adapt dynamically as the user moves. Safety alerts shall always be prioritized above all other system outputs, and normal navigation shall operate in a hands-free manner.

For object interaction, the system shall detect interactive objects such as door handles and elevator panels, estimate their spatial location, track the user’s hand in real time, and guide hand

movement using directional haptic feedback. Confirmation feedback shall be provided once correct alignment is achieved.

The system shall support recognition of text, signs, and known individuals, converting recognized information into speech. Facial recognition functionality shall operate only with explicit user consent and shall associate identities with user-defined names.

Automatic mode switching shall manage transitions between navigation, interaction, OCR, and hand guidance modes while preventing unstable or rapid switching. Users shall retain the ability to manually override modes using voice commands.

All sensitive data shall be processed locally, encrypted at rest, and manageable by the user. The system shall support degraded operation when sensors fail, revert to safe navigation modes during critical failures, and immediately notify the user of any safety-impacting errors.

4.3 User Requirements

From the user's perspective, Echora shall enable safe and independent navigation in indoor and outdoor environments while wearing the smart glasses. Users shall receive continuous awareness of obstacles and open paths through intuitive audio and haptic feedback and be immediately alerted to dangerous situations such as stairs or approaching obstacles. The system shall allow natural movement without requiring the user to hold or aim a device.

Users shall be able to locate and interact with common objects such as door handles and elevator buttons with minimal trial and error. Hand guidance shall be precise, with clear confirmation feedback when alignment is correct.

The system shall provide access to environmental information including text, signs, and familiar individuals through audio output. Interaction shall be primarily automatic, supplemented by simple voice commands, and feedback shall remain clear, non-overlapping, and interruptible at any time.

Comfort, trust, and safety are essential. The smart glasses shall remain comfortable for long-term wear, feedback shall not overwhelm the user, and the system shall always prioritize safety while handling personal and biometric data securely and privately.

4.4 Non-Functional Requirements

Non-functional requirements address performance, security, reliability, safety, usability, maintainability, and operational constraints.

The system shall meet strict real-time performance targets, including frame processing latency below 40 milliseconds, navigation feedback latency below 120 milliseconds, and danger alert latency below 100 milliseconds. Mode switching shall occur within 200 milliseconds, and audio and haptic outputs shall execute promptly without overlap.

Security requirements mandate on-device processing, strong encryption of biometric data using AES-256, secure device pairing, and strict user consent for sensitive features. Privacy-by-design principles shall govern all data handling.

The system shall achieve high reliability with a mean time between failures exceeding 10,000 hours and support graceful degradation and fallback modes. Availability during active use shall exceed 99.5%.

Safety requirements include limits on audio volume and haptic intensity, thermal and electrical safety compliance, fail-safe behavior, and adherence to medical device and electromagnetic compatibility standards.

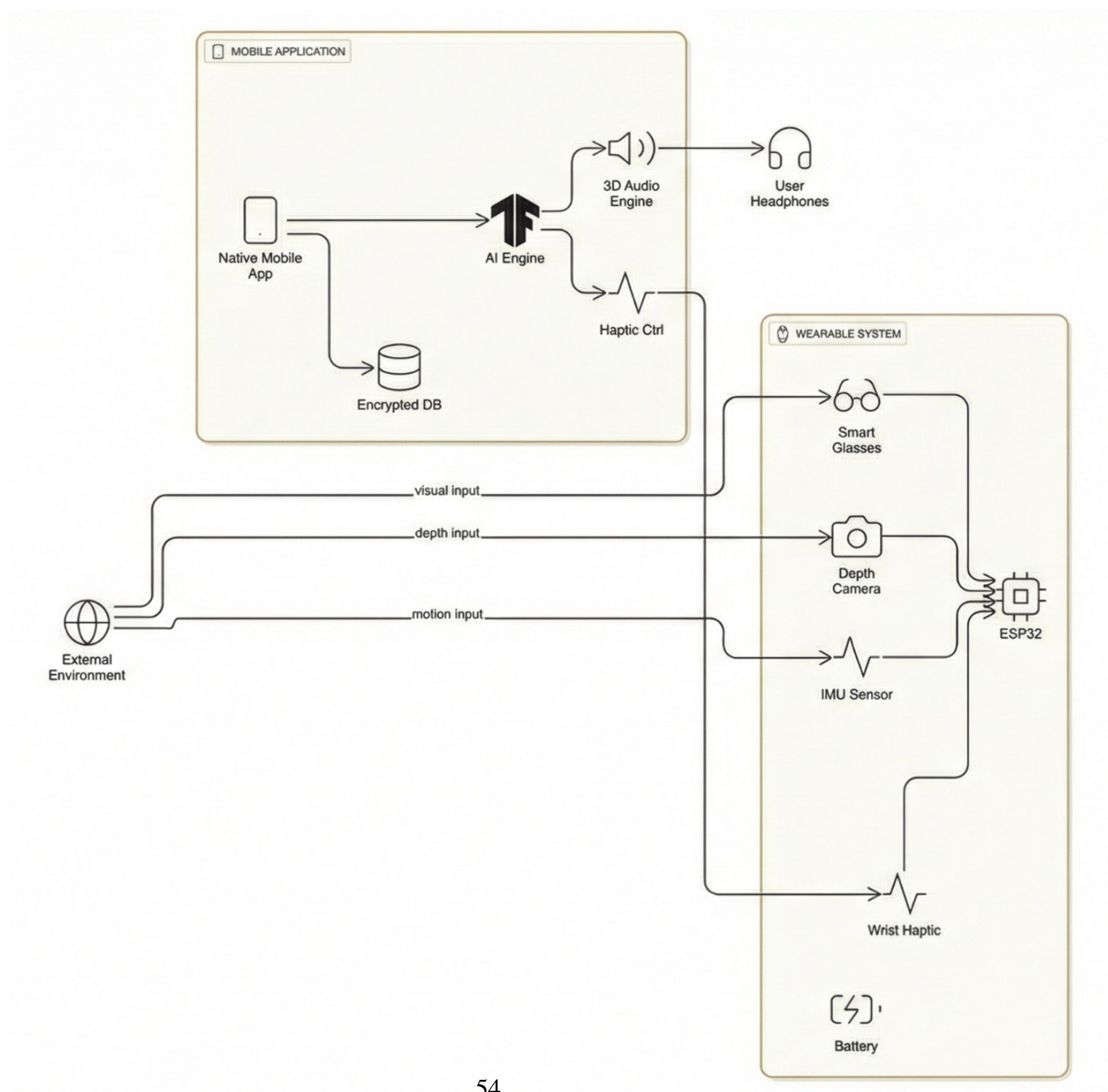
Usability requirements emphasize learnability, efficiency, accessibility compliance, voice control, and consistent multi-modal feedback. Maintainability requirements include modular design, over-the-air updates, diagnostic support, and long-term hardware serviceability.

Operational requirements specify deployment procedures, environmental operating conditions, regulatory compliance, and end-of-life considerations including data migration, recycling, and long-term support commitments.

Chapter 5

System Design

5.1 System Architecture Diagram



5.2 Proposed Model Diagram

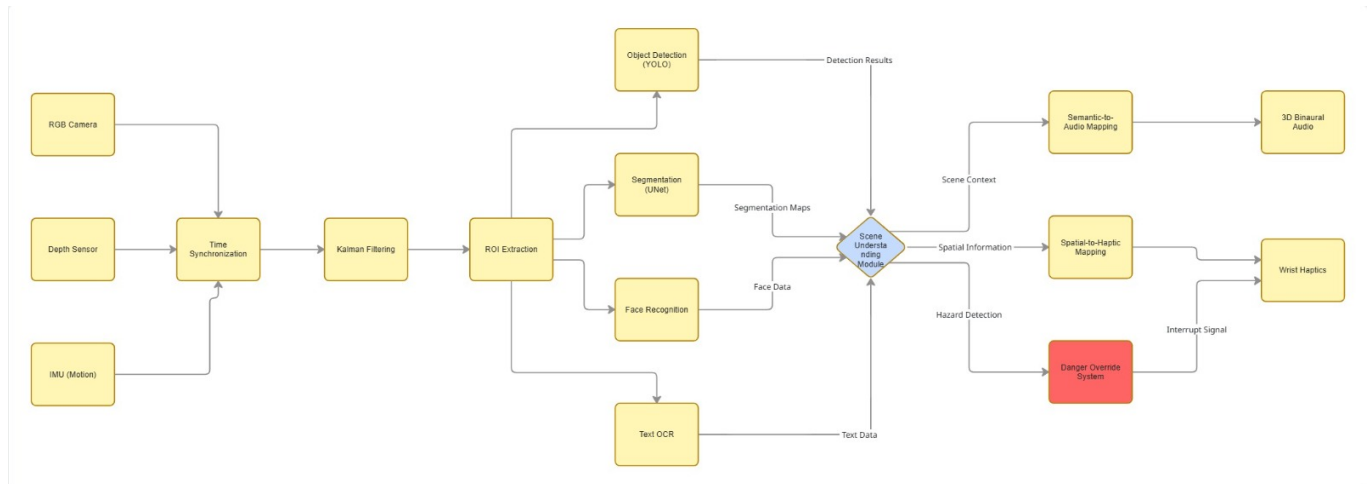


Figure 5.2: Proposed Model Diagram

5.3 AI Pipeline Diagram

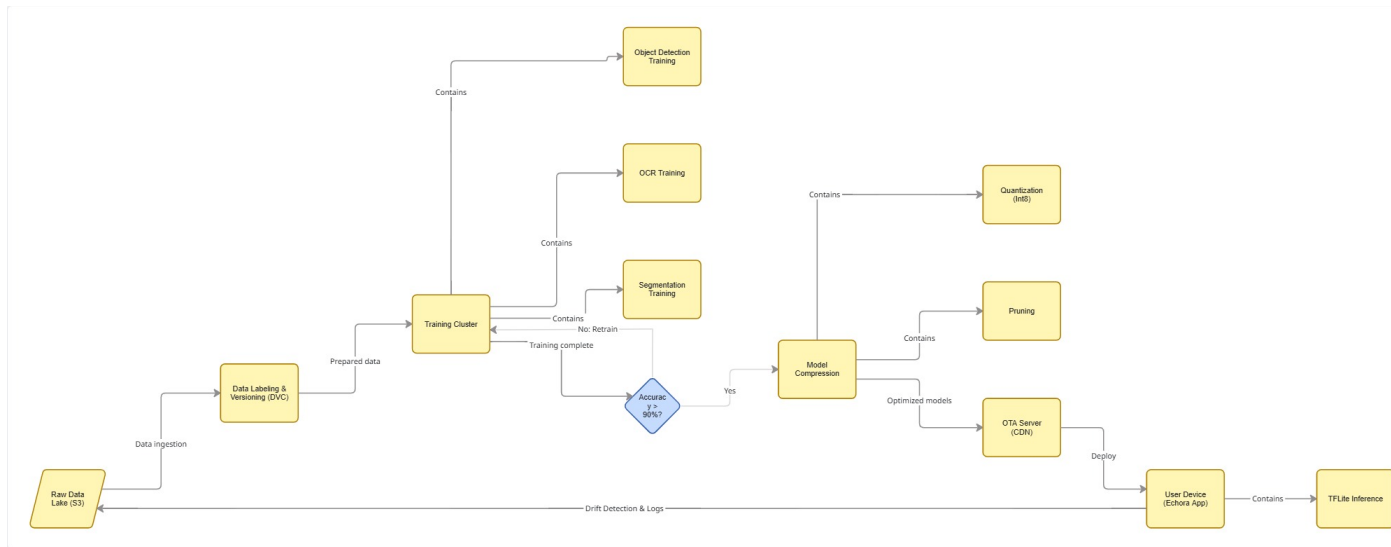


Figure 5.3: AI Pipeline Diagram

5.4 Data Flow Diagrams

5.4.1 Level 0

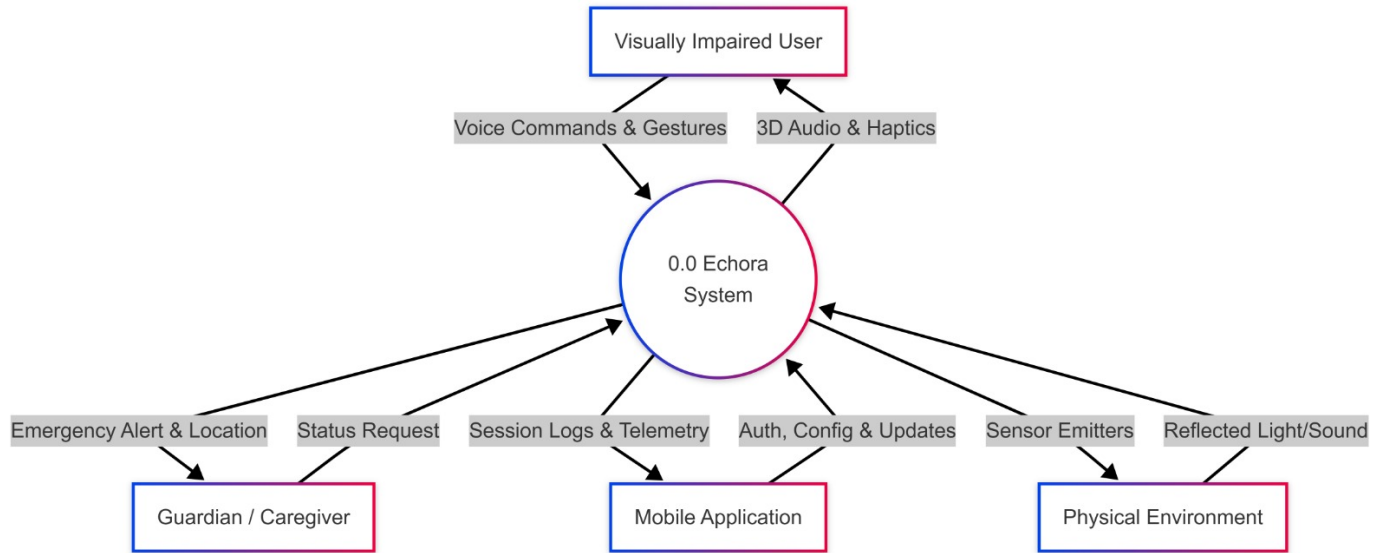


Figure 5.4: Data Flow Diagram Level 0

5.4.2 Level 1

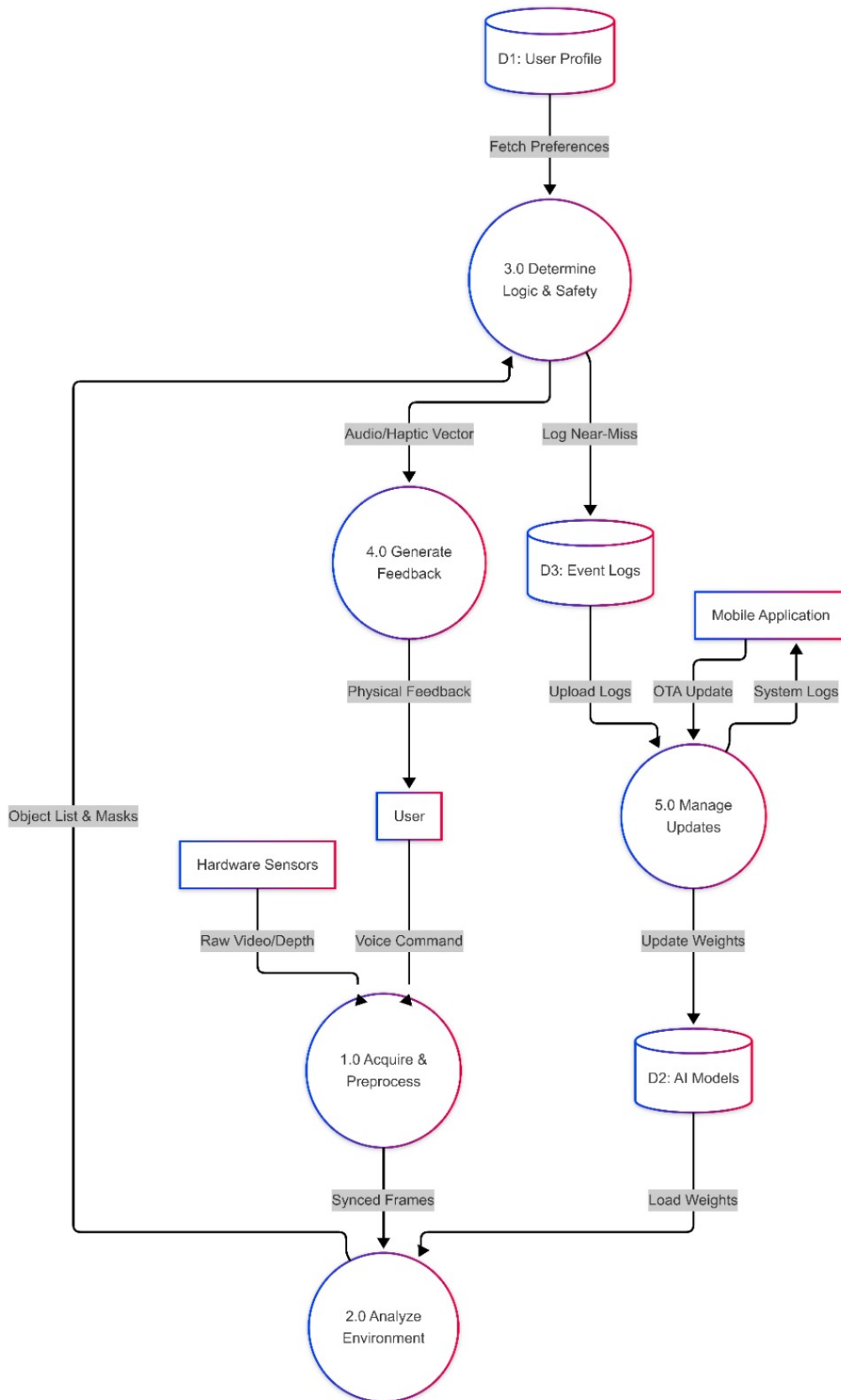


Figure 5.5: Data Flow Diagram Level 1

5.4.3 Level 2

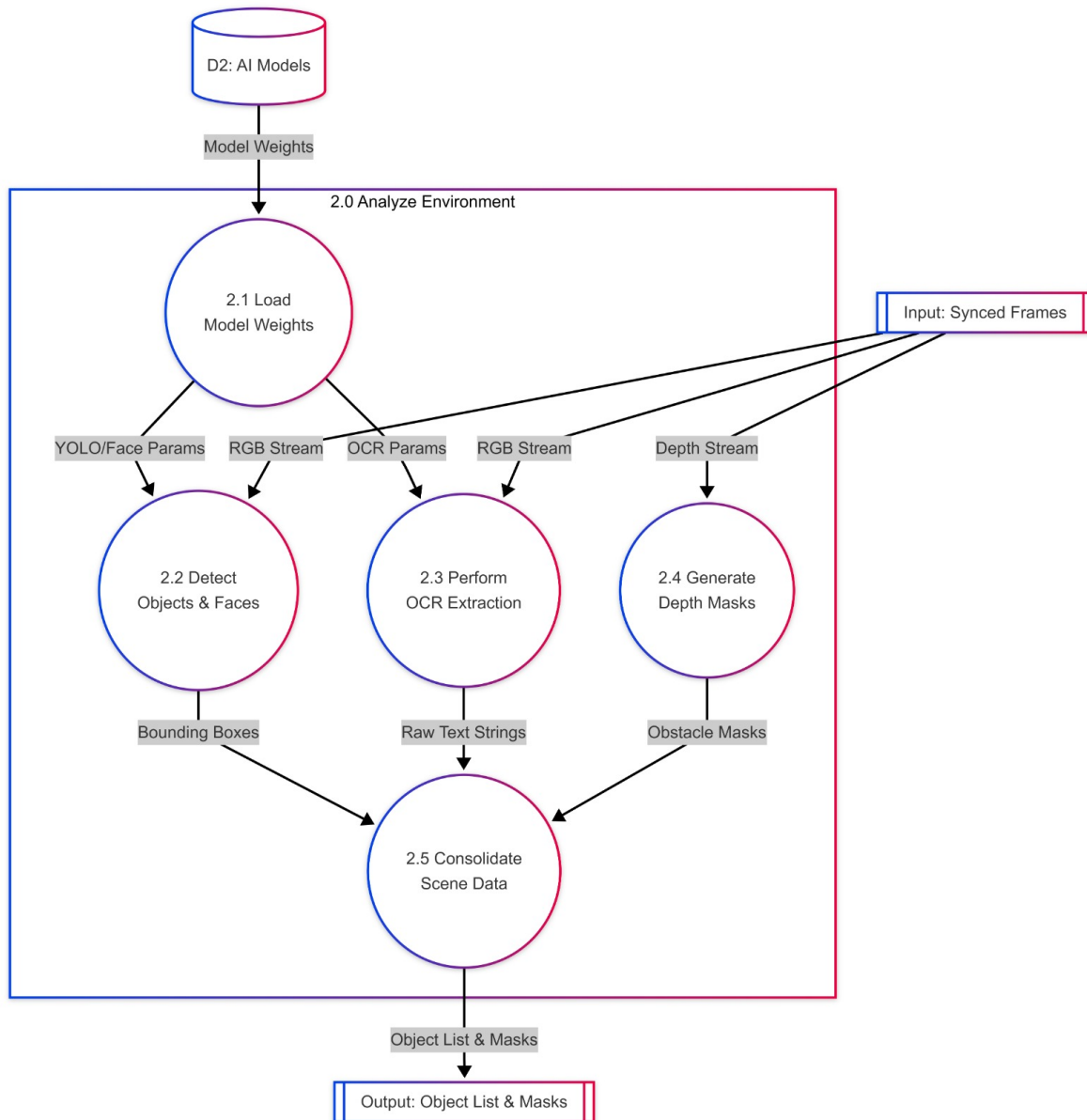


Figure 5.6: Data Flow Diagram Level 2

5.5 Sequence Diagrams

5.5.1 Text Reading (OCR) & Speech

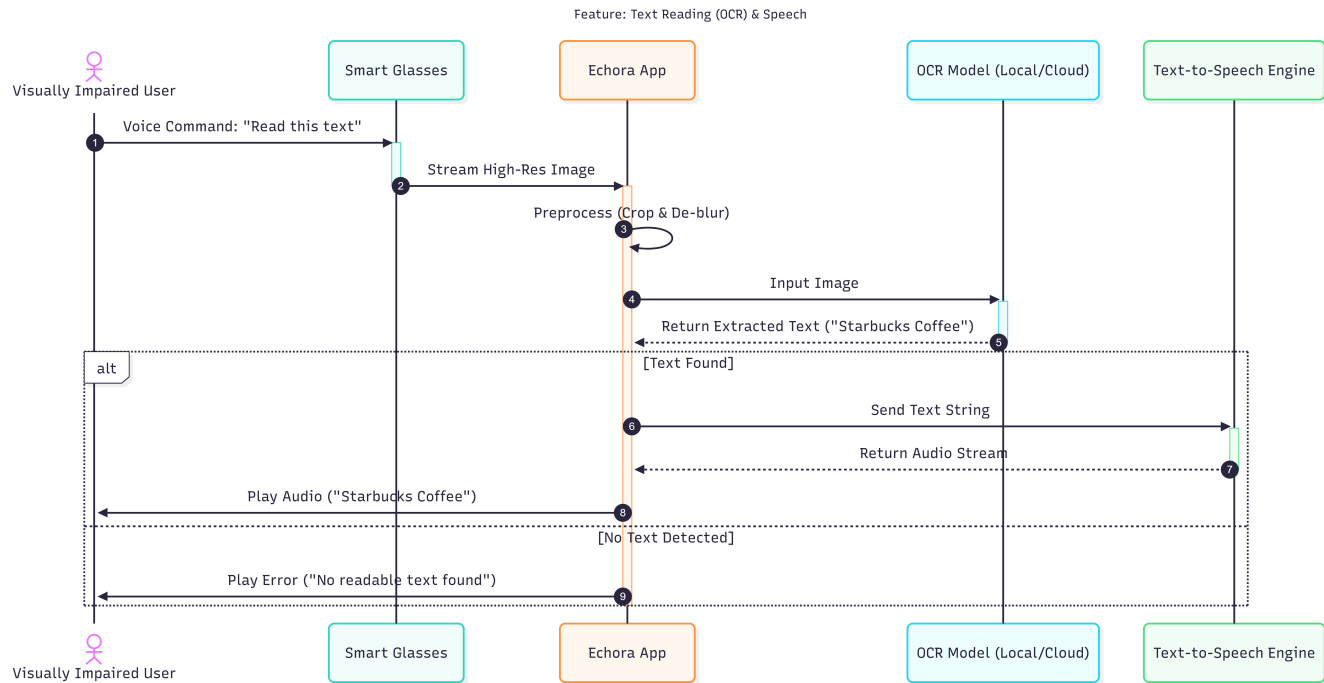


Figure 5.7: Text Reading (OCR) & Speech Sequence Diagram

5.5.2 Emergency Panic Mode

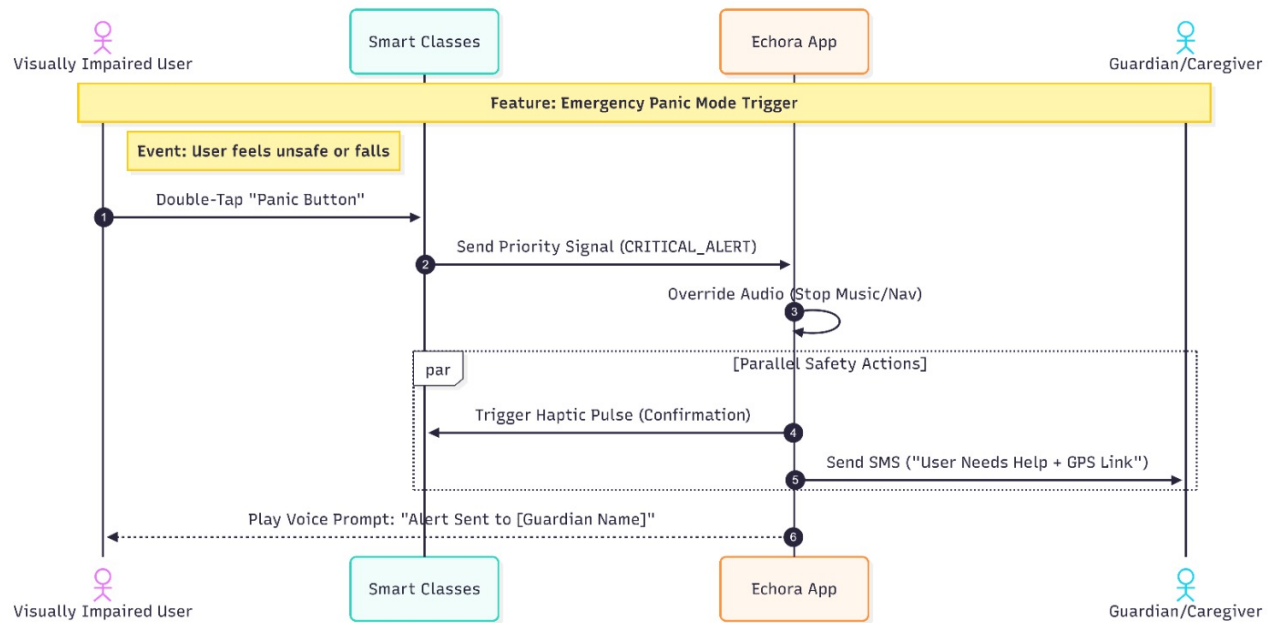


Figure 5.8: Emergency Panic Mode Trigger Sequence Diagram

5.5.3 Object Interaction & Hand Guidance

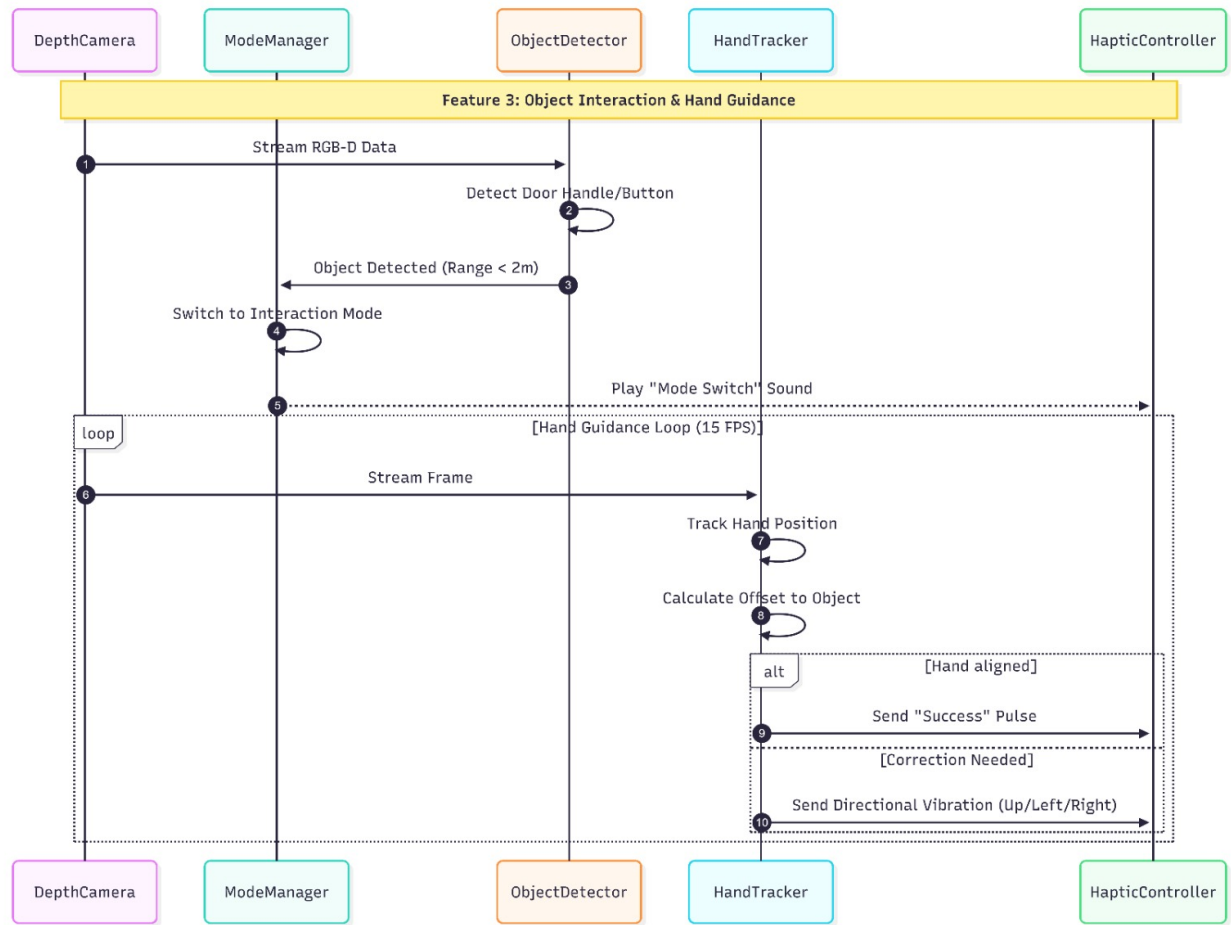


Figure 5.9: Object Interaction & Hand Guidance Sequence Diagram

5.6 Use Cases Diagrams

5.6.1 Use Case Diagram

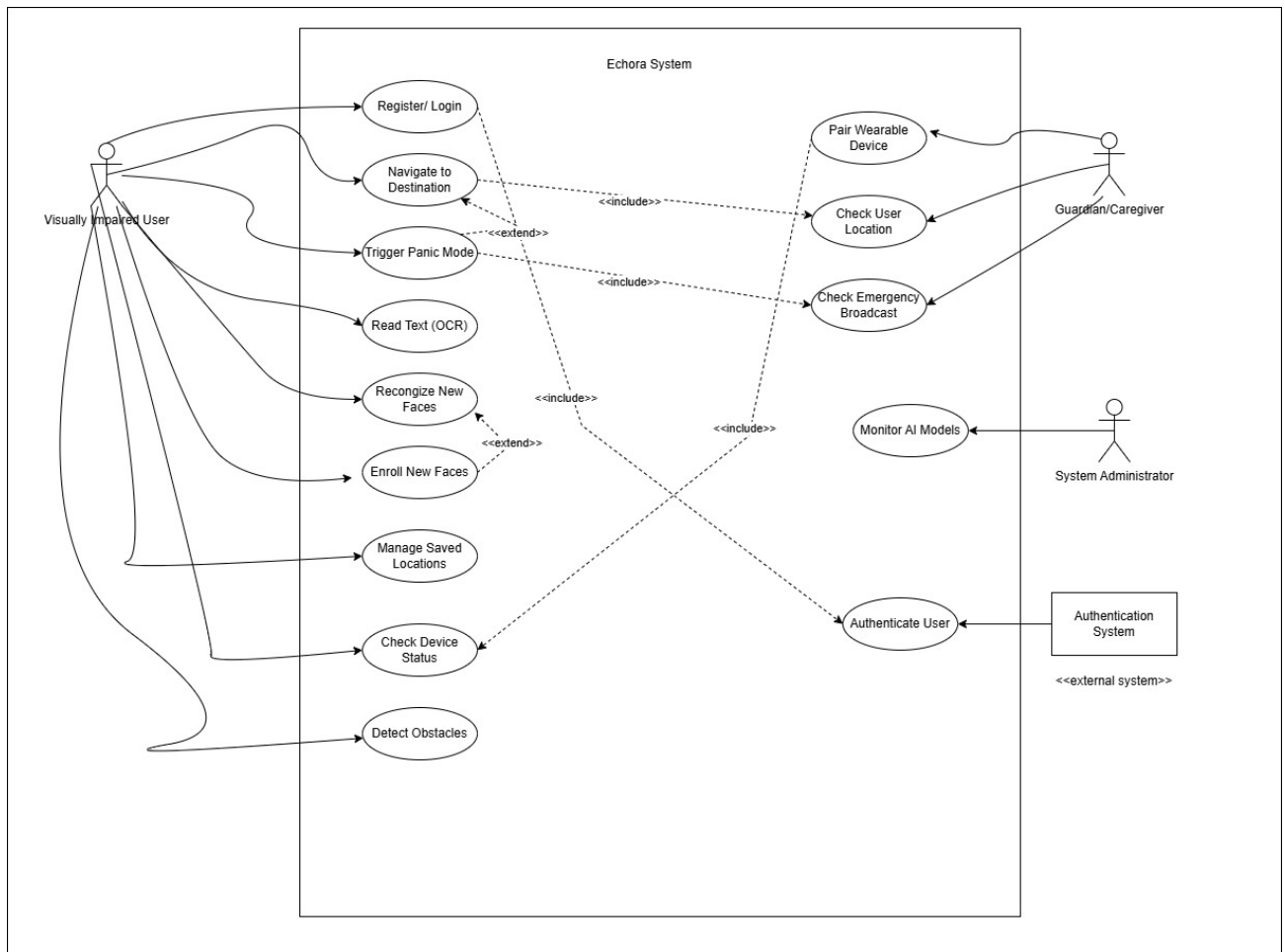


Figure 5.10: Use Case Diagram

5.7 Class Diagram

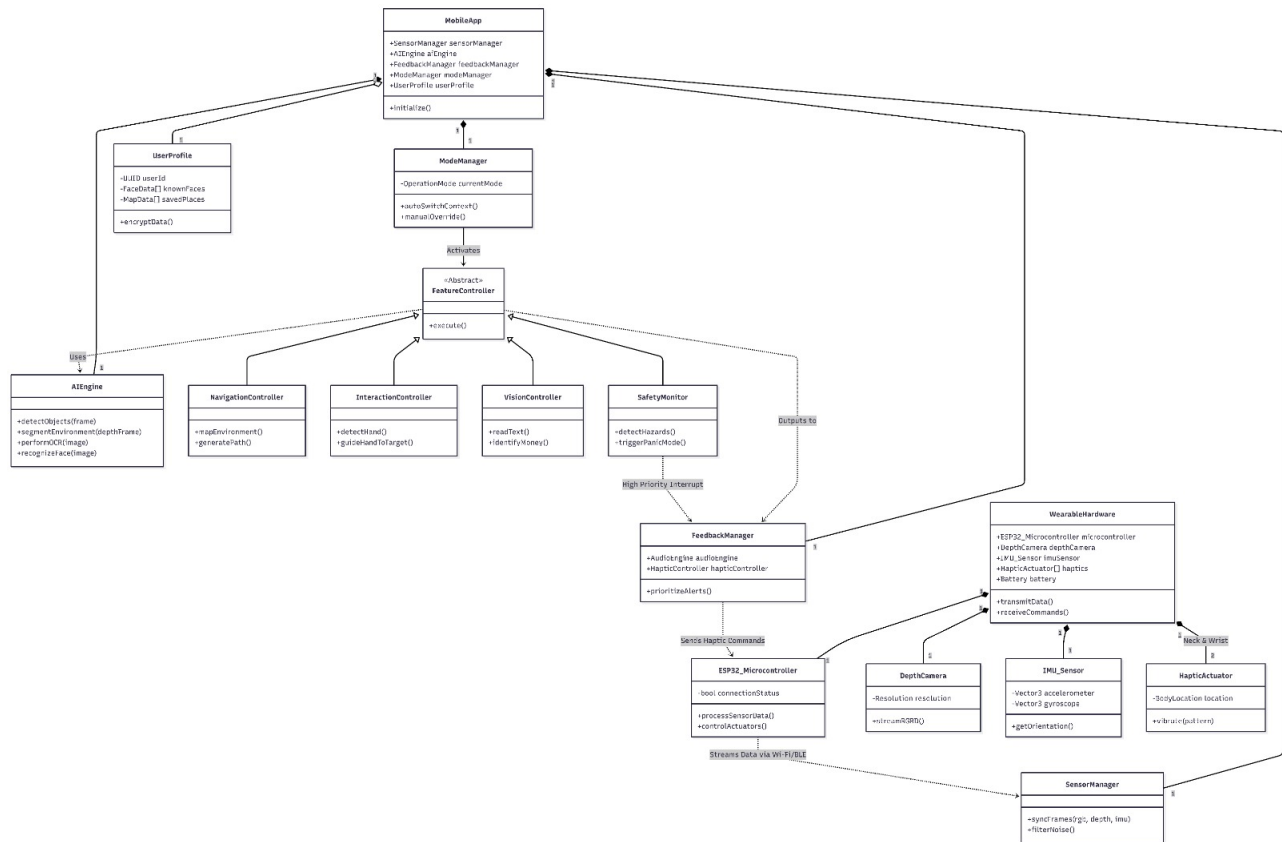


Figure 5.11: Class Diagram

5.8 Entity Relationship Diagram (ERD)

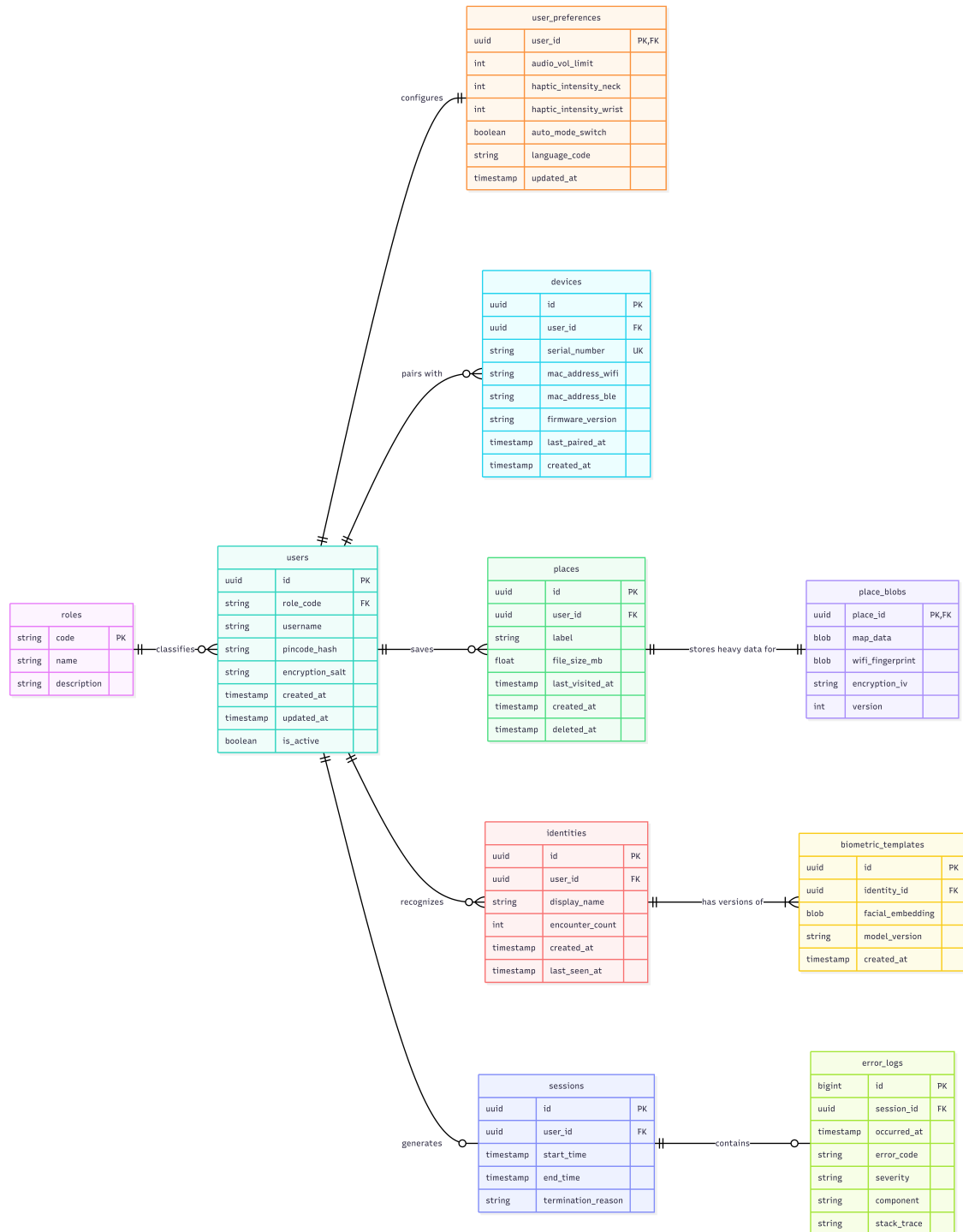


Figure 5.12: Entity Relationship Diagram

Chapter 6

Implementation

6.1 Introduction

This chapter presents the implementation details of the Echora system, describing how the proposed design and requirements are realized in a practical, working prototype. It focuses on the datasets, development environment, core algorithms, and system integration strategies used to transform the conceptual architecture into an operational AI-powered sensory substitution system.

The implementation emphasizes real-time performance, modularity, and on-device processing to meet the strict latency, privacy, and reliability requirements defined in earlier chapters. All computationally intensive tasks, including computer vision, depth processing, sensor fusion, and machine learning inference, are executed locally on the mobile device, ensuring independence from cloud connectivity and preserving user privacy.

This chapter begins by describing the datasets used to train and evaluate the system's machine learning models. It then outlines the development environment and software tools employed, followed by an overview of the application prototype and its core algorithms. Finally, the chapter discusses system integration challenges and implementation considerations encountered during development, highlighting design decisions made to ensure robustness and usability in real-world environments.

Together, the contents of this chapter demonstrate how Echora's theoretical framework and system design are translated into a functional assistive solution capable of supporting navigation, object interaction, and information recognition for Blind and Low Vision users.

6.2 Datasets

The Echora system relies on multiple publicly available datasets obtained from Kaggle, each selected to support a specific functional component of the AI-powered sensory substitution pipeline. These datasets collectively provide visual, depth, textual, and semantic information required for navigation, object interaction, and information recognition tasks. The use of diverse datasets enables robust training, validation, and evaluation of machine learning models under varied real-world conditions.

Indoor Environment and Depth Perception Datasets

To support real-time environment mapping and obstacle detection, Echora utilizes RGB-D datasets that provide paired color and depth information:

- **NYU Depth V2 Dataset (Kaggle Mirror):** This dataset contains RGB and depth images of indoor environments such as corridors, rooms, and offices. It is widely used for indoor

scene understanding, depth-based segmentation, and spatial layout estimation. The depth annotations enable accurate distance measurement and obstacle proximity estimation, which are essential for continuous environment mapping and audio-based navigation.

- **SUN RGB-D Dataset (Kaggle Version):** The SUN RGB-D dataset includes RGB-D images with labeled objects such as walls, furniture, doors, and household items. It supports multi-class object detection and semantic segmentation in indoor environments, allowing Echora to distinguish between navigable spaces and obstacles.

Object Detection and Interaction Datasets

For detecting and classifying interactive objects such as doors, handles, furniture, and people, the following datasets are used:

- **COCO 2017 (Common Objects in Context):** The COCO dataset provides large-scale object annotations for everyday objects including chairs, tables, doors, and humans. It is used for pretraining lightweight object detection models such as YOLO and MobileNet, which are later fine-tuned for Echora's specific interaction scenarios.
- **Open Images Dataset V6 (Kaggle):** This dataset offers a wide variety of object classes with bounding-box annotations, enabling improved generalization and robustness in object detection, particularly in complex indoor and semi-structured environments.

Hand Detection and Interaction Datasets

Hand tracking is a core component of Echora's object interaction and hand guidance system:

- **EgoHands Dataset (Kaggle Mirror):** EgoHands provides first-person perspective images with annotated hand regions. This dataset is well-suited for wearable-camera scenarios and supports accurate hand detection and tracking during close-range interactions.
- **Hand Gesture Recognition Dataset:** This dataset contains labeled hand poses and gestures, enabling the development and evaluation of hand presence and orientation detection models used in guiding the user's hand toward interaction targets.

Text and Sign Recognition Datasets

To enable text reading and sign detection, Echora incorporates scene text datasets that reflect real-world visual conditions:

- **ICDAR 2015 Scene Text Dataset (Kaggle Version):** This dataset includes images of natural scenes containing text with bounding-box annotations. It supports training text detection models capable of identifying text regions under varying lighting and orientation conditions.
- **SynthText Dataset (Kaggle Mirror):** SynthText provides a large volume of synthetically generated text over real images, allowing effective pretraining of text detection networks before fine-tuning on real-world data.

Currency Recognition Datasets

For banknote and coin identification, the following datasets are employed:

- **Currency Recognition Dataset (Kaggle):** This dataset contains images of banknotes from multiple currencies and denominations. It is used to train classification models that recognize monetary values based on visual patterns, color, and layout.
- **Coin Classification Dataset (Kaggle):** This dataset includes labeled images of coins captured under different orientations and lighting conditions, supporting accurate coin value identification.

Facial Recognition Dataset

For known individual recognition, a standard benchmark dataset is used strictly for feature extraction and evaluation:

- **Labeled Faces in the Wild (LFW) Dataset (Kaggle):** LFW contains face images collected under unconstrained conditions and is commonly used to train and evaluate facial embedding models. In Echora, this dataset is used only to develop the underlying recognition model, while all real user data is collected locally with explicit user consent.

Overall, the combination of large-scale public datasets and targeted custom data provides a solid foundation for training and evaluating Echora's machine learning components. The diversity and scale of these datasets enable robust model generalization, effective feature learning, and reliable real-time performance in assistive navigation and interaction scenarios.

6.3 Development Environment

The Echora system was implemented using a mobile-centric architecture in which the smartphone serves as the primary processing unit. The mobile application was developed using the *Flutter* framework, enabling a single cross-platform codebase that supports both Android and iOS devices while maintaining near-native performance.

Flutter was selected due to its high rendering efficiency, strong accessibility support, and seamless integration with native modules required for real-time sensor processing and machine learning inference. The application logic was written in the Dart programming language, while performance-critical components were implemented using platform-specific native code when necessary.

The development environment consisted of Android Studio and Visual Studio Code as integrated development environments (IDEs). Version control was managed using Git, enabling collaborative development and iterative testing.

All artificial intelligence and computer vision processing were executed locally on the mobile device using TensorFlow Lite and MediaPipe frameworks. OpenCV was utilized for image preprocessing and depth data refinement. On-device text-to-speech (TTS) and speech-to-text (STT) APIs were employed to provide auditory feedback and voice-based user interaction.

Wireless communication between the wearable hardware and the mobile application was achieved using Wi-Fi Direct for high-bandwidth sensor data streaming and Bluetooth Low Energy (BLE) for low-latency haptic feedback commands. This configuration ensured real-time responsiveness while maintaining power efficiency and data privacy.

6.4 Application Prototype

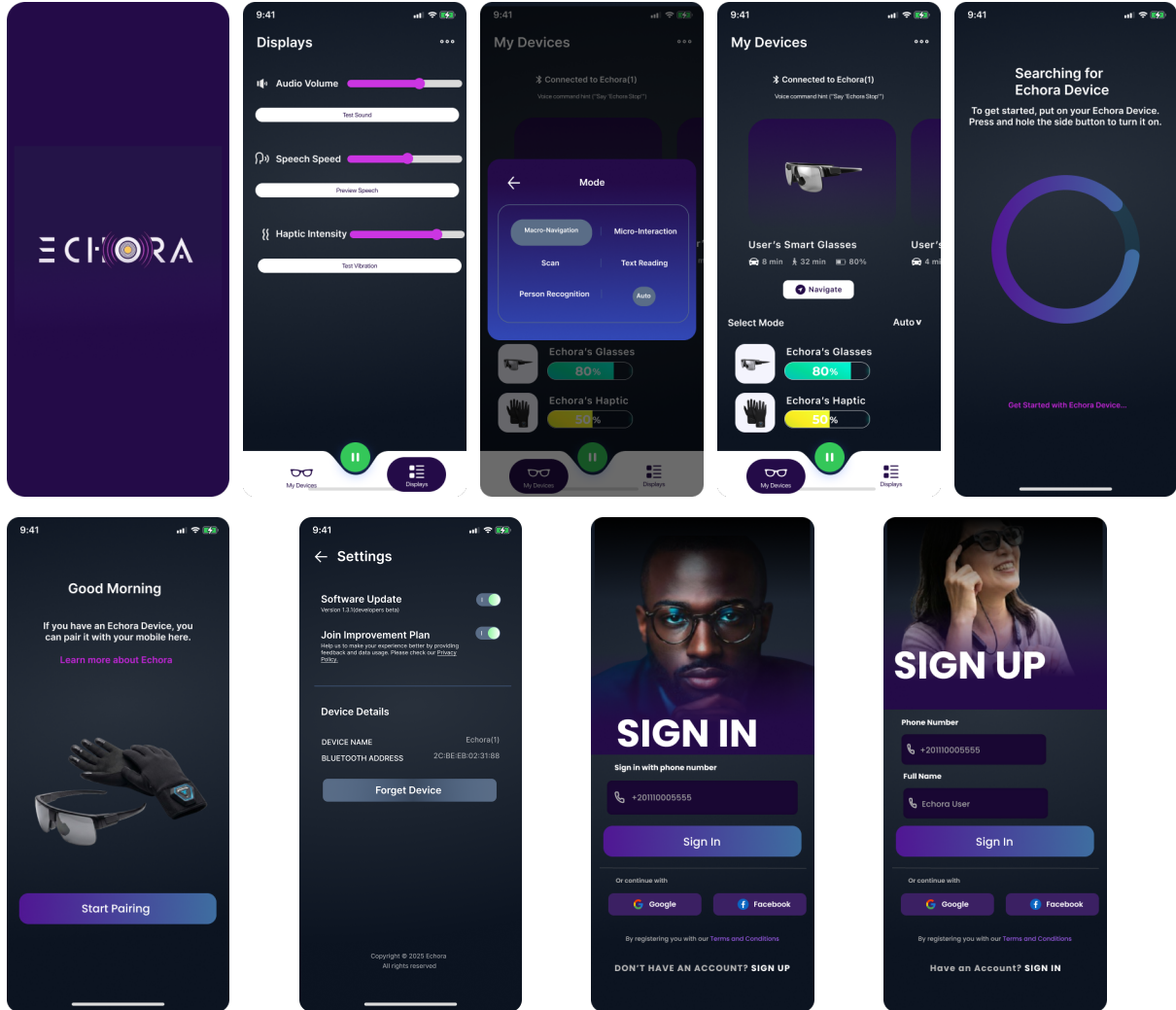


Figure 6.1: Echora Mobile Application Prototype Screens

6.5 Core Algorithms and Models

The Echora system integrates multiple lightweight artificial intelligence models optimized for real-time execution on mobile devices.

For environmental navigation, depth frames are fused with inertial measurement unit (IMU) data using a sensor fusion algorithm based on Kalman filtering. A lightweight convolutional neural network (CNN) performs real-time segmentation of walls, obstacles, humans, and open pathways. The resulting spatial information is translated into continuous spatial audio cues, where distance, direction, and urgency are encoded through volume, stereo positioning, and pitch variation.

Object interaction and hand guidance are implemented using an object detection model based on MobileNet or YOLO architectures. Detected interactive objects such as door handles and elevator panels are localized using depth measurements. Hand tracking is performed using MediaPipe Hands at a minimum of 15 frames per second. Directional correction vectors are continuously computed and translated into wrist-mounted haptic feedback until proper alignment is achieved.

Text and sign recognition is achieved through a two-stage pipeline consisting of scene text detection using EAST or DBNet models, followed by optical character recognition using a mobile-optimized Tesseract engine. Recognized text is converted into speech using an on-device TTS engine.

Critical danger detection employs time-to-collision estimation using depth difference kernels and optical flow techniques. Sudden terrain changes, such as stairs or drop-offs, are identified through ground plane discontinuity analysis. These alerts override all other system outputs to ensure immediate user safety.

6.6 Implementation Challenges

Several challenges were encountered during the implementation of the Echora system. Achieving real-time performance on mobile hardware while running multiple AI models simultaneously required careful optimization, dynamic resolution scaling, and adaptive fallback strategies.

Synchronizing heterogeneous sensor streams, including RGB, depth, and IMU data, proved challenging due to differing sampling rates and communication latencies. Timestamp-based buffering and alignment were implemented to maintain spatial consistency.

Designing effective audio and haptic feedback without overwhelming the user required extensive prioritization logic and mode management to reduce cognitive load. Additionally, mobile resource constraints such as battery consumption and thermal throttling necessitated aggressive power management and computational efficiency.

Environmental variability, including lighting changes, motion blur, and partial occlusions, also affected perception accuracy. These issues were mitigated through confidence thresholds, graceful degradation, and fallback modes.

6.7 Conclusion

This chapter presented the implementation of the Echora system using a Flutter-based mobile application as the core processing platform. The described development environment, application prototype, and AI-driven perception pipeline demonstrate the feasibility of delivering real-time, on-device sensory substitution for Blind and Low Vision users.

Despite challenges related to performance optimization, sensor reliability, and feedback design, the implemented prototype successfully integrates navigation assistance, object interaction, text recognition, and safety monitoring within a unified framework. The results validate the proposed system architecture and confirm that modern smartphones can effectively support advanced assistive technologies without reliance on cloud-based processing.

The implemented system provides a strong foundation for future enhancements, extended user testing, and potential real-world deployment.

Chapter 7

Conclusion and the Future Work

This project presented the design and implementation of *Echora*, a mobile-centered assistive system aimed at enhancing environmental perception and independent navigation for Blind and Low Vision (BLV) users. The system was developed with a strong emphasis on real-time performance, privacy preservation, and usability in everyday environments.

Echora integrates RGB-D vision, inertial sensing, and on-device artificial intelligence to provide continuous spatial awareness, object interaction assistance, text recognition, and safety alerts. All perception and decision-making processes are executed locally on a smartphone, eliminating reliance on cloud-based services and ensuring low latency, robustness, and user data privacy. Haptic feedback is delivered exclusively through a wrist-mounted actuator, while spatial audio cues complement tactile feedback to reduce cognitive load and improve situational understanding.

The system architecture was carefully designed to balance functionality with hardware constraints, leveraging lightweight machine learning models optimized for mobile deployment. A Flutter-based cross-platform mobile application was developed to unify sensing, perception, feedback generation, and user interaction within a single, accessible interface. The implemented prototype demonstrates that modern mobile devices are capable of supporting advanced assistive technologies without the need for specialized external infrastructure.

Throughout development, several challenges were encountered, including real-time performance optimization, sensor synchronization, and feedback prioritization. These challenges were addressed through efficient model selection, modular system design, and adaptive feedback strategies. The resulting system validates the feasibility of delivering multimodal sensory substitution using vision-based perception and wearable haptic feedback.

In conclusion, Echora demonstrates a practical and scalable approach to assistive technology that enhances autonomy and safety for BLV users. The project establishes a solid foundation for future work, including extended user evaluations, usability studies, and incremental feature expansion, while maintaining its core principles of on-device processing, simplicity, and user-centered design.